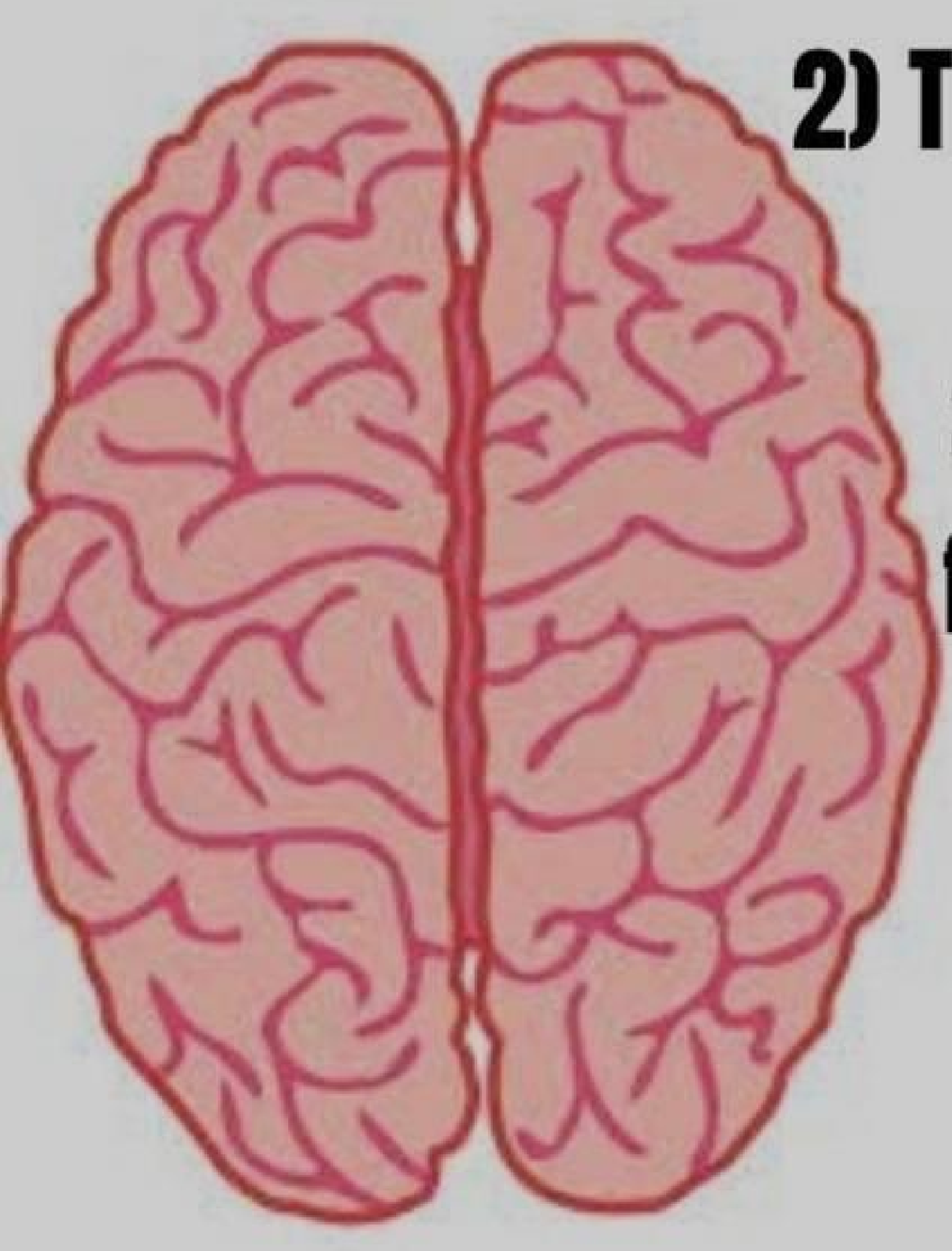
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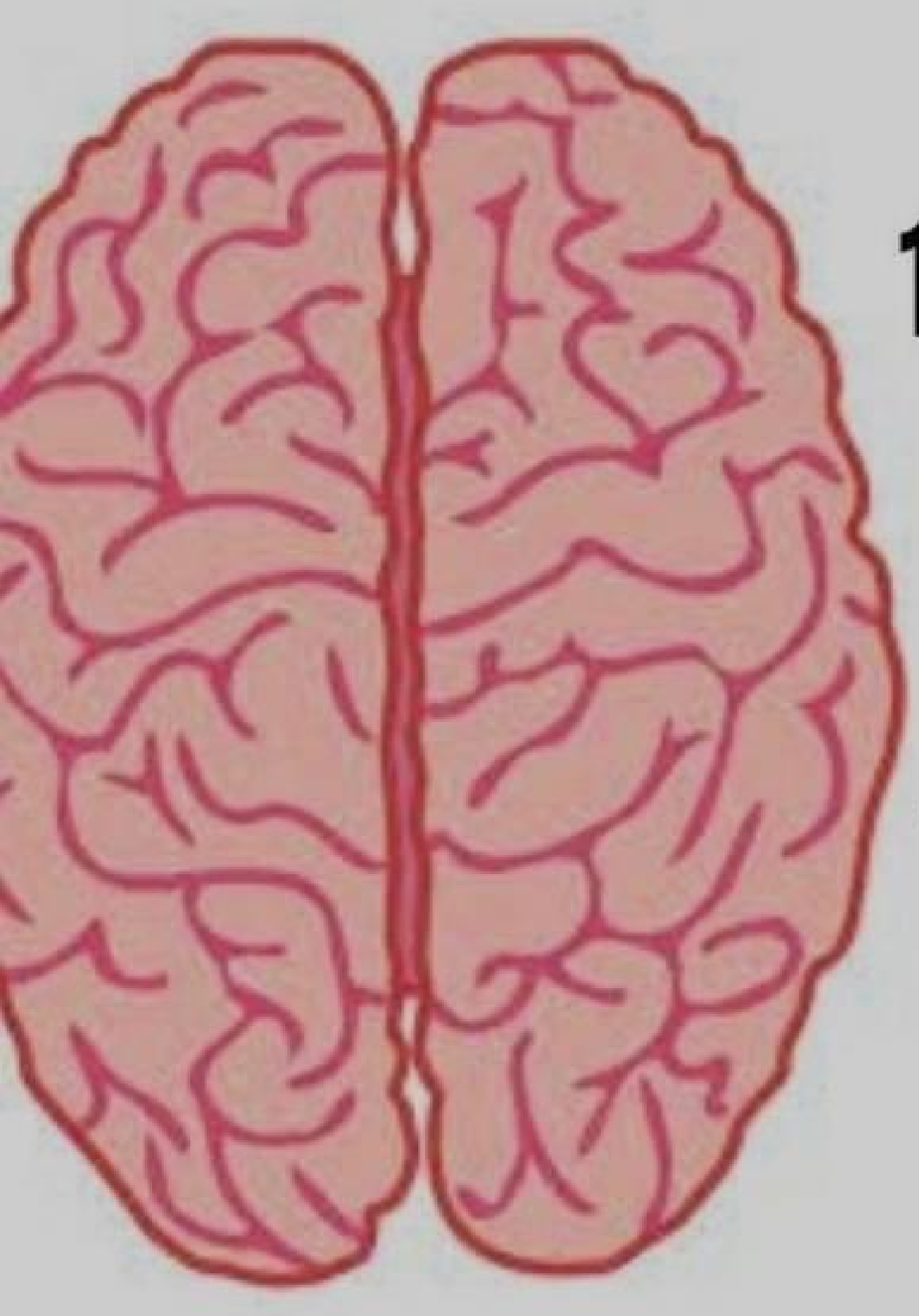
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2) T.S Medulla Oblongata (3 layers)
& T.S. Midbrain (2 Layers)
3)Superorlateral surface and functional area of Cerebrum
4) Cerebellum Anatomy
5) Thalamus Anatomy
6) Circle of Willis
7) Spinal cord blood supply
8) Lateral & Third ventricle
9) Floor of Fourth ventricle

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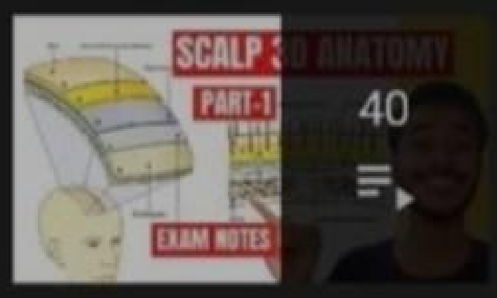
PART-2



10) Basal Nuclei
11) Internal capsule
12) Corpus callosum
13) White matter of cerebrum
14) Cristerna
15) Optic nerve
16) Glossopharyngeal nerve
17) facial nerve
18) vagus nerve
19) Brown sequard syndrome
20) Medial medullary syndrome

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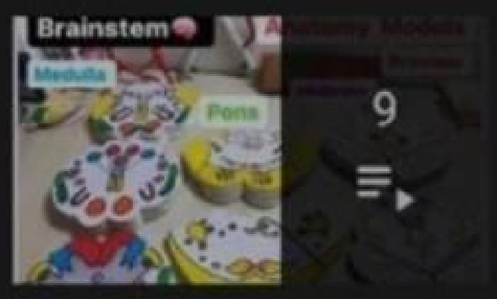
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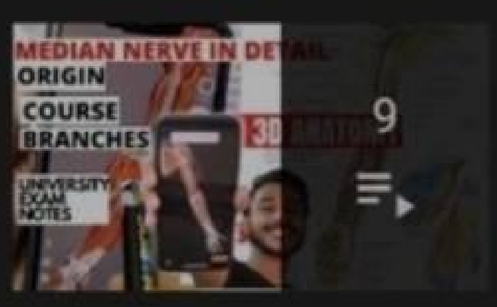
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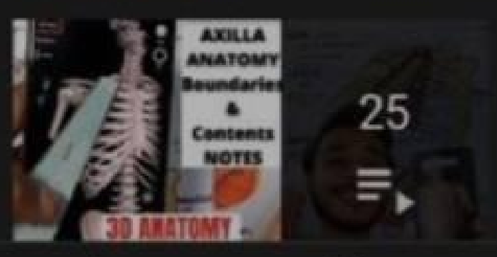
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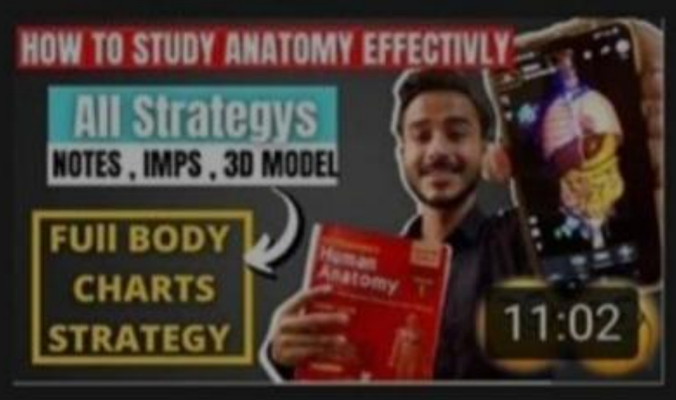


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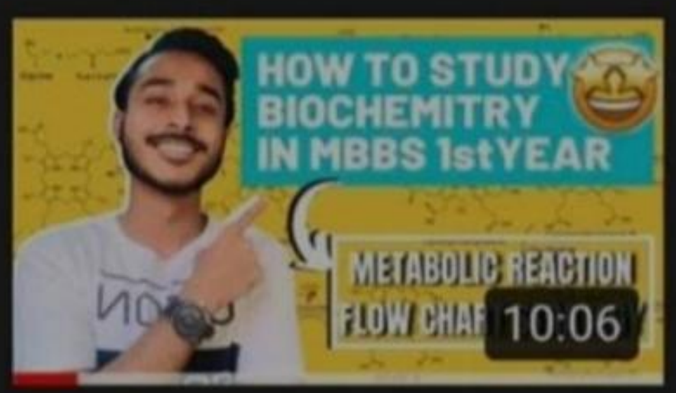
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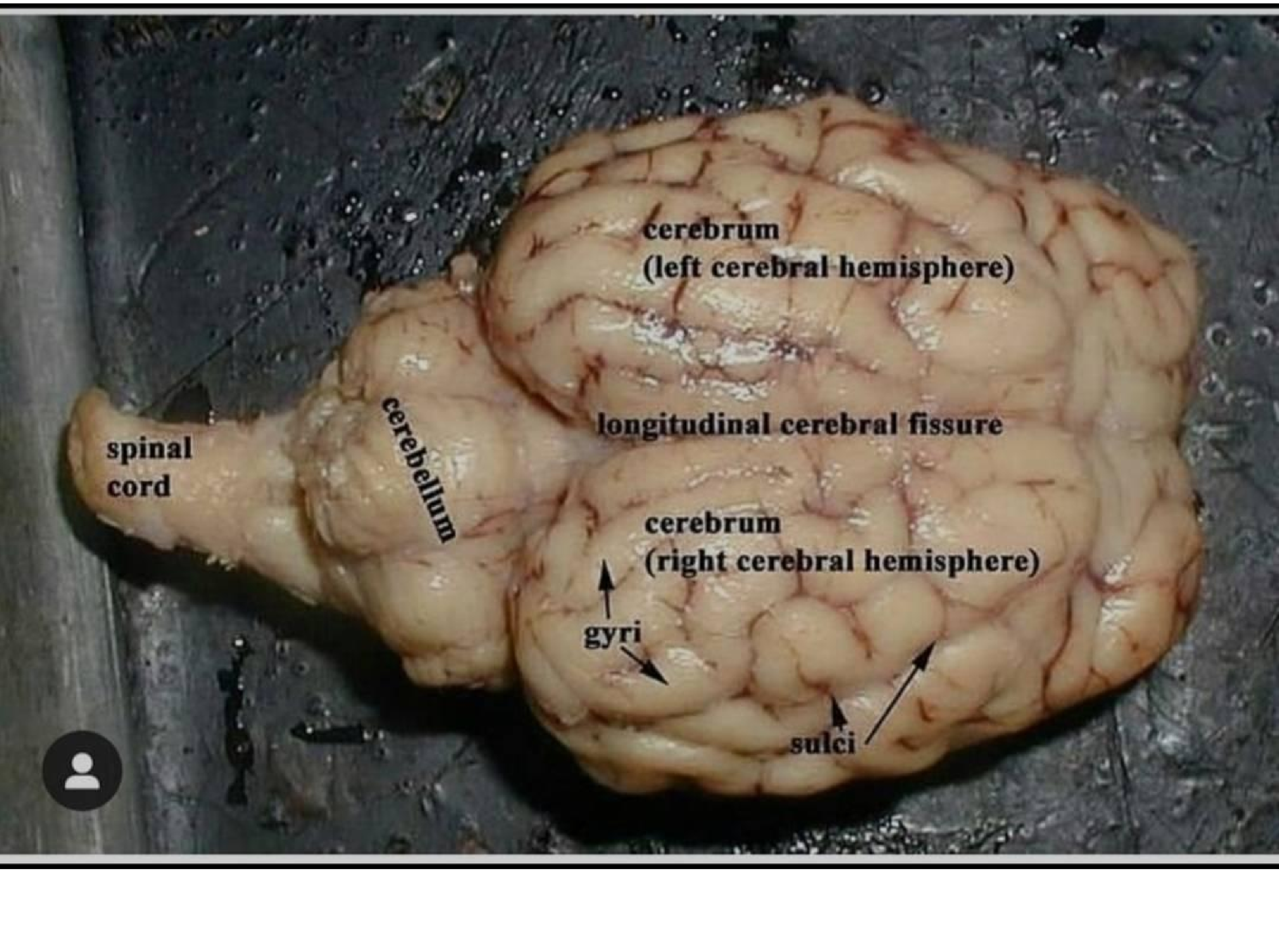


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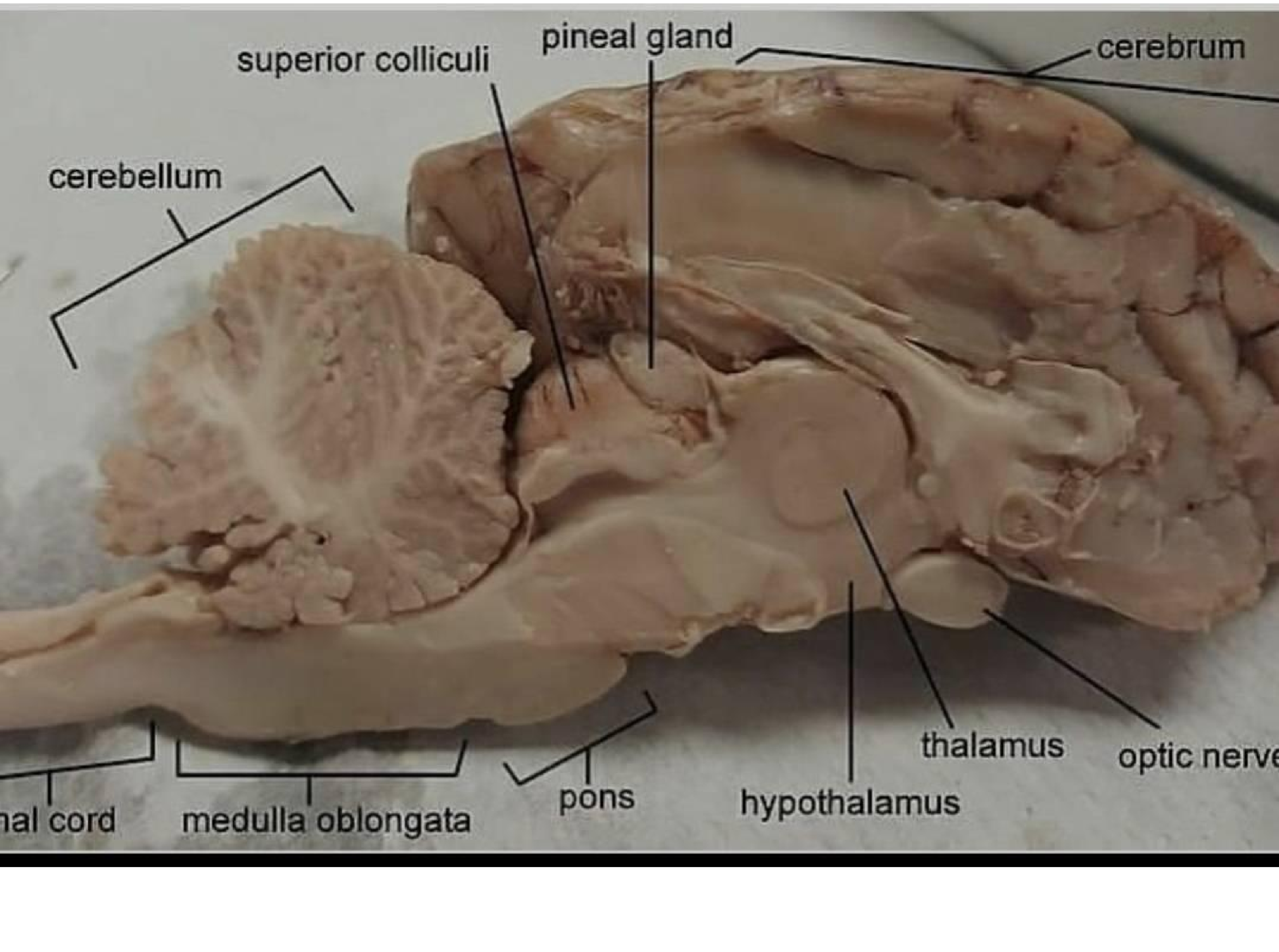


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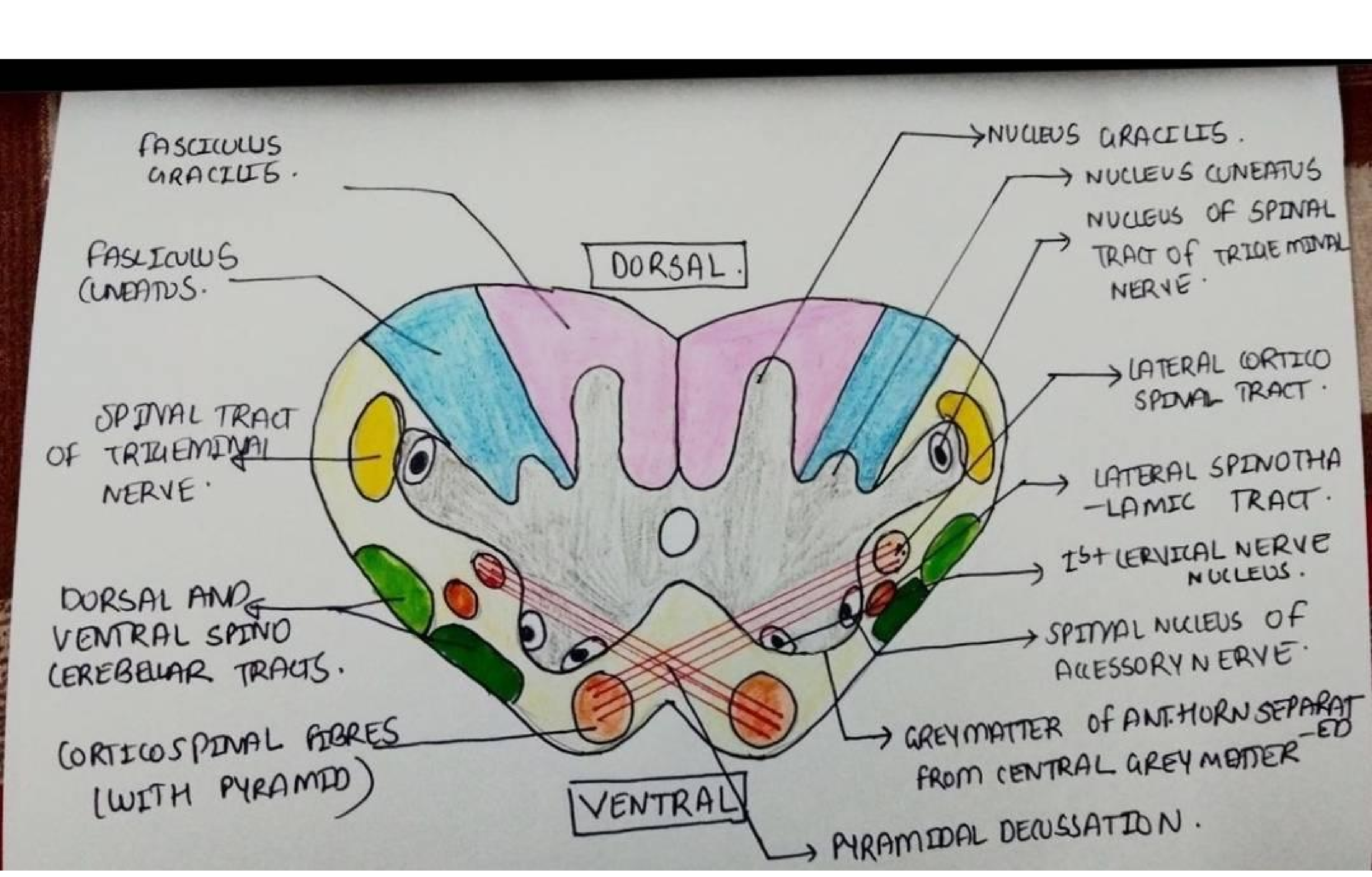
Ascending tracts

Descending tracts



SPINAL CORD

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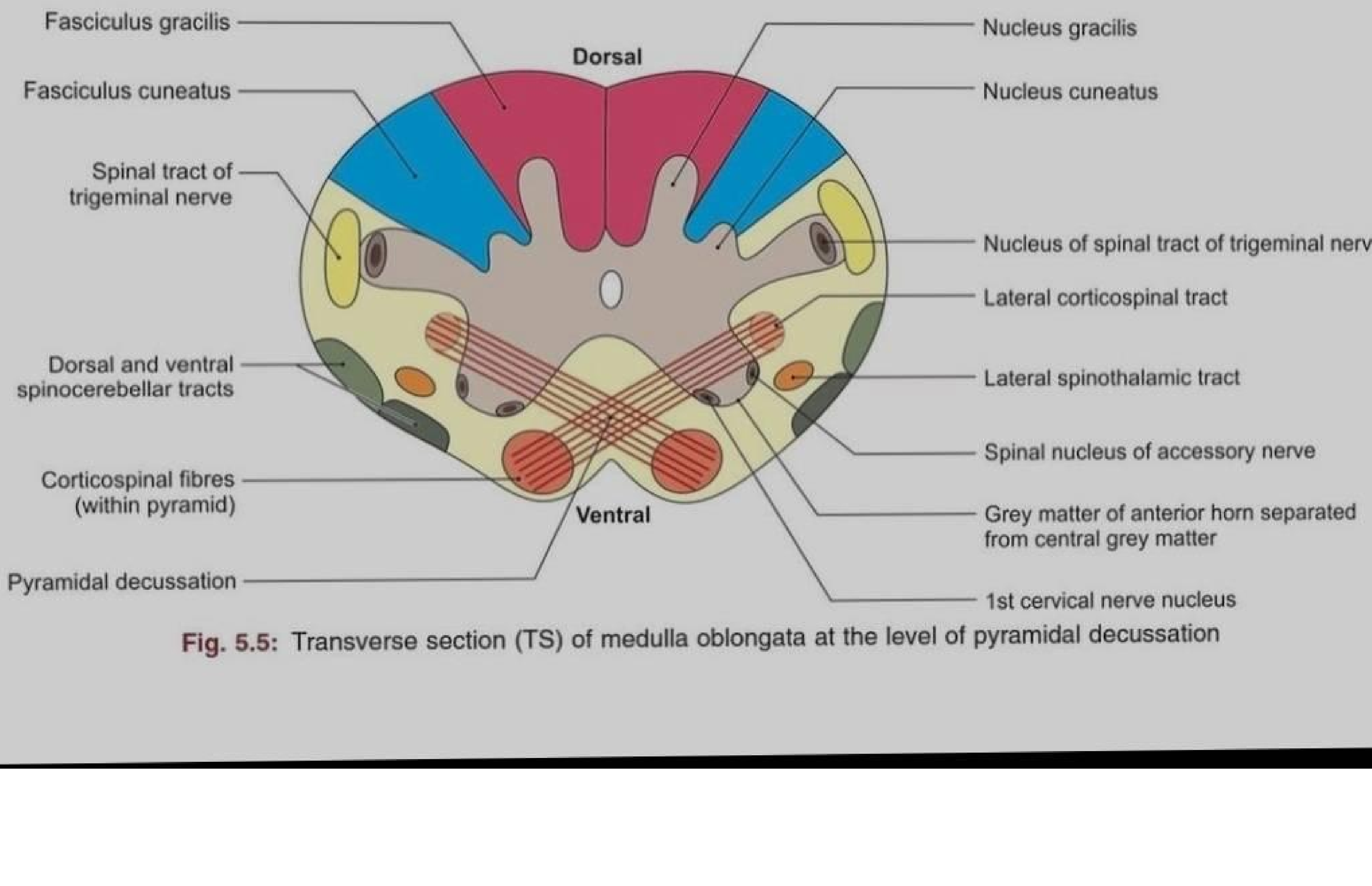
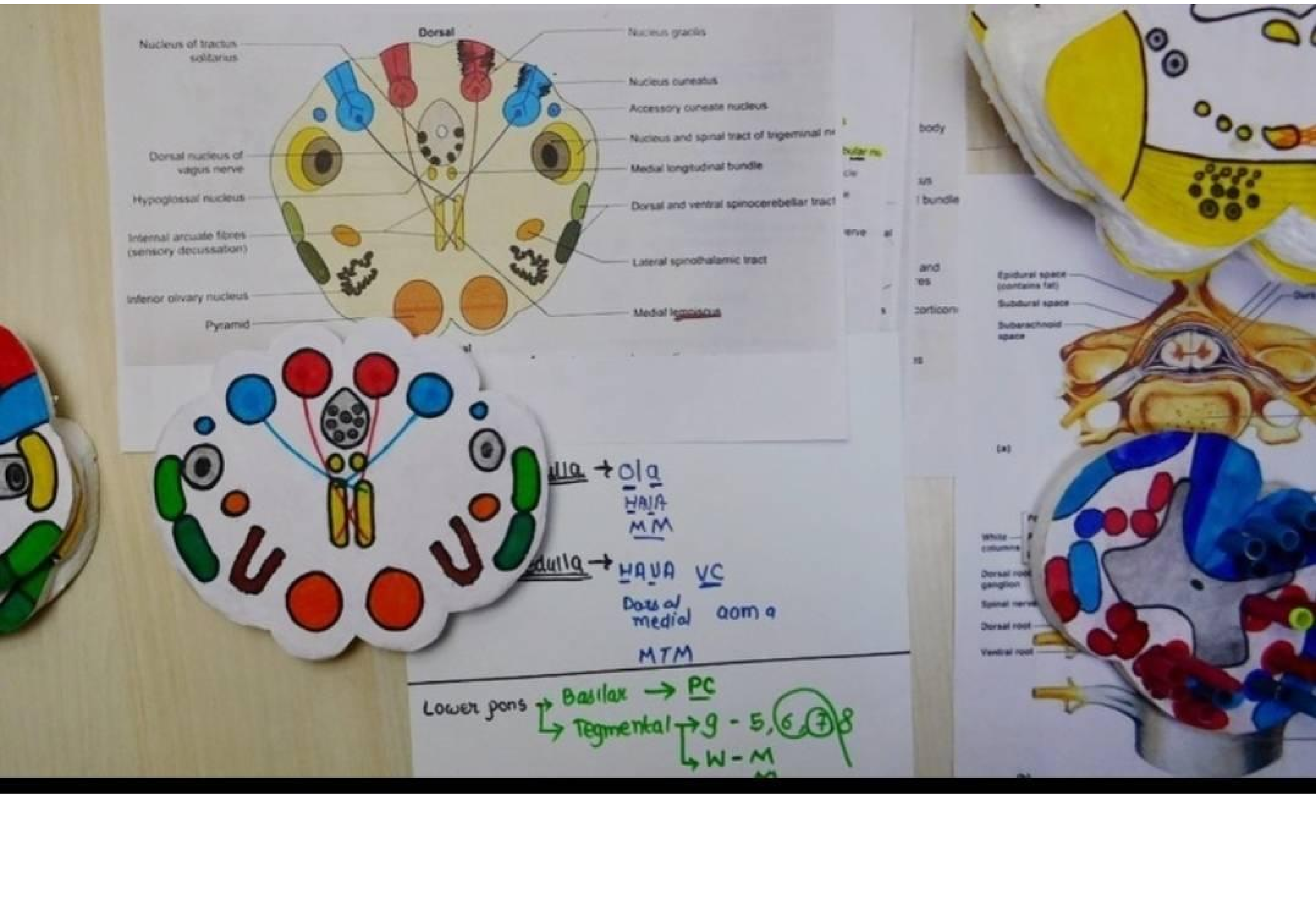


Fig. 5.5: Transverse section (TS) of medulla oblongata at the level of pyramidal decussation

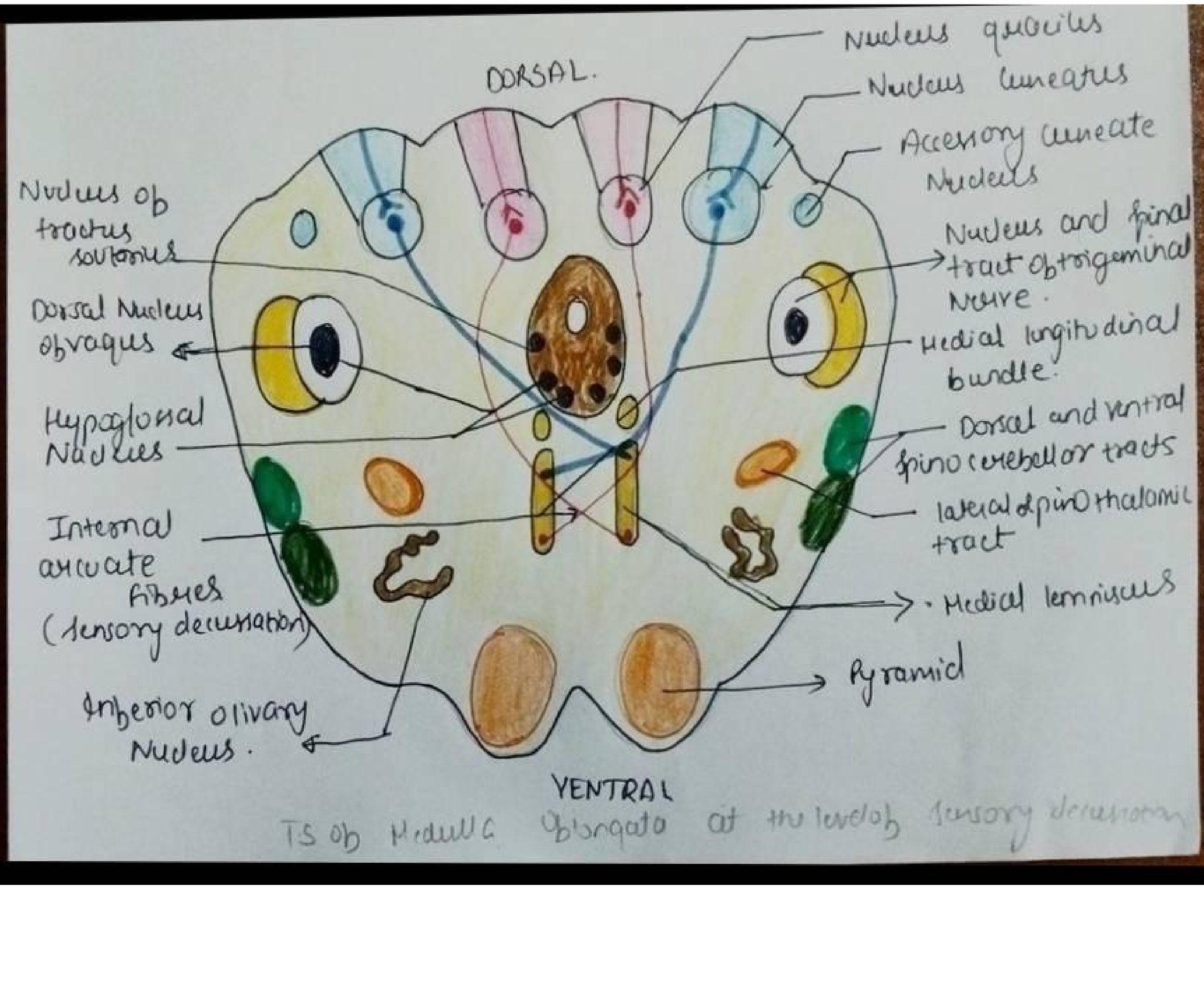
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TS of Medulla Oblongata at the level of sensory decussation

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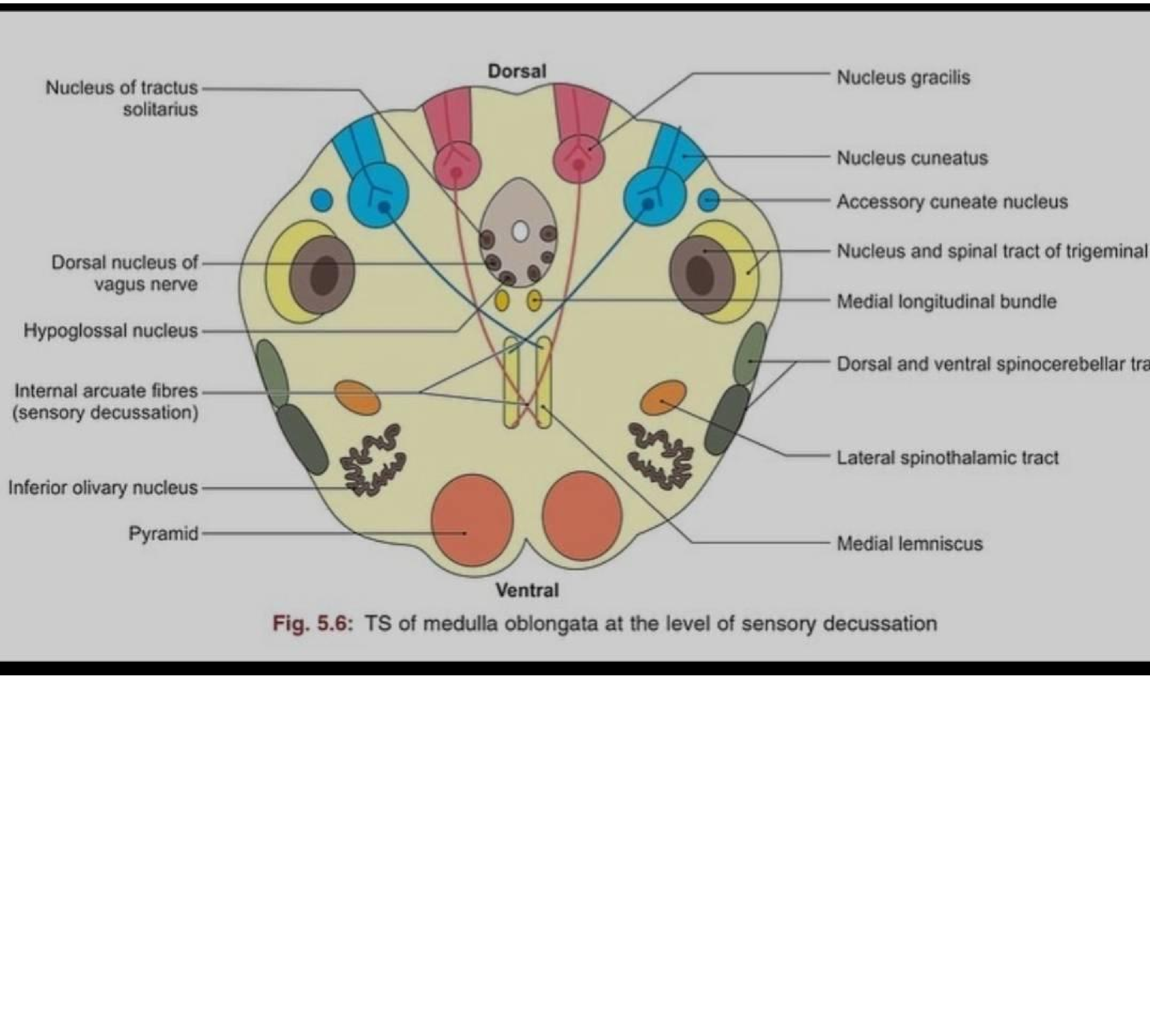
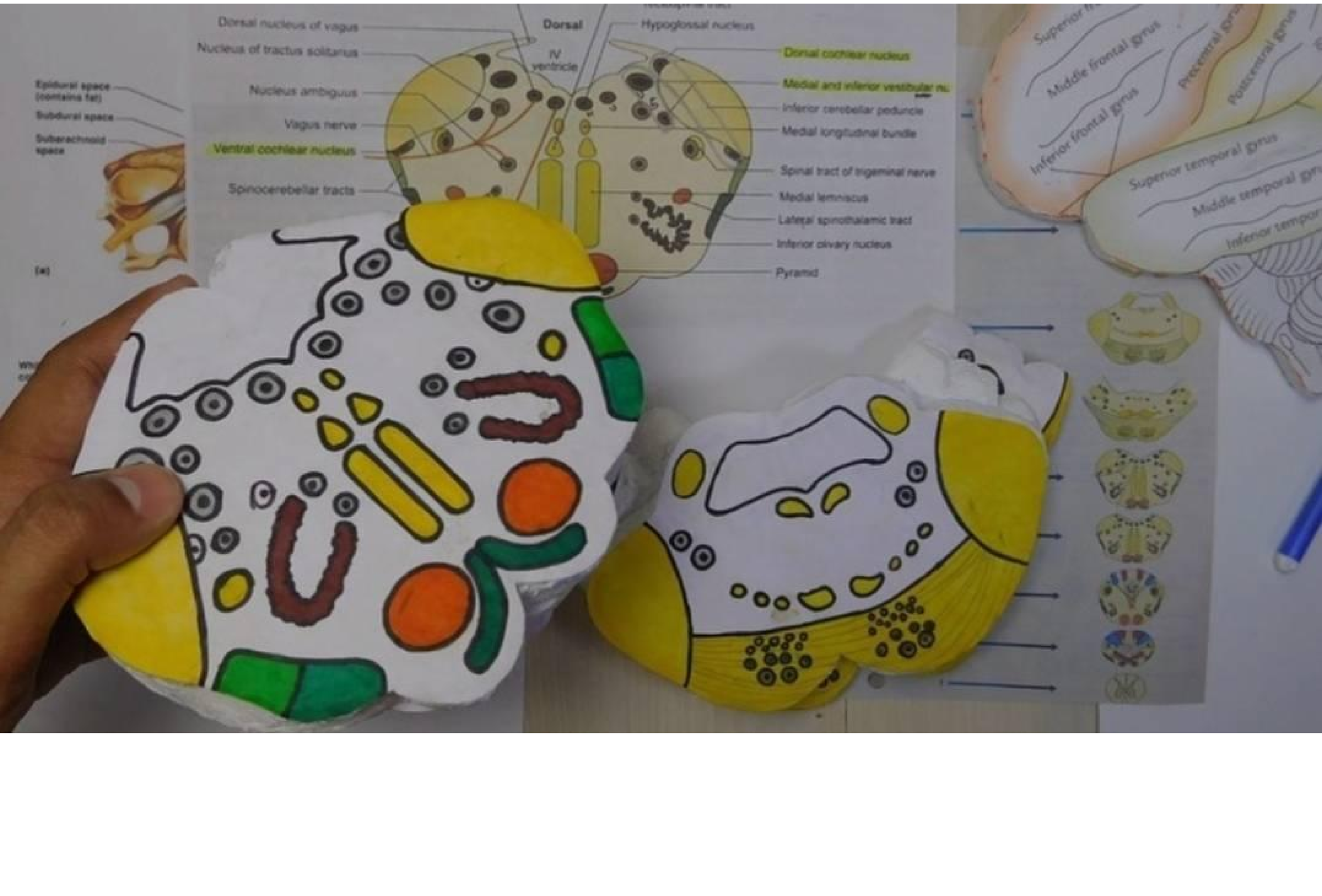


Fig. 5.6: TS of medulla oblongata at the level of sensory decussation

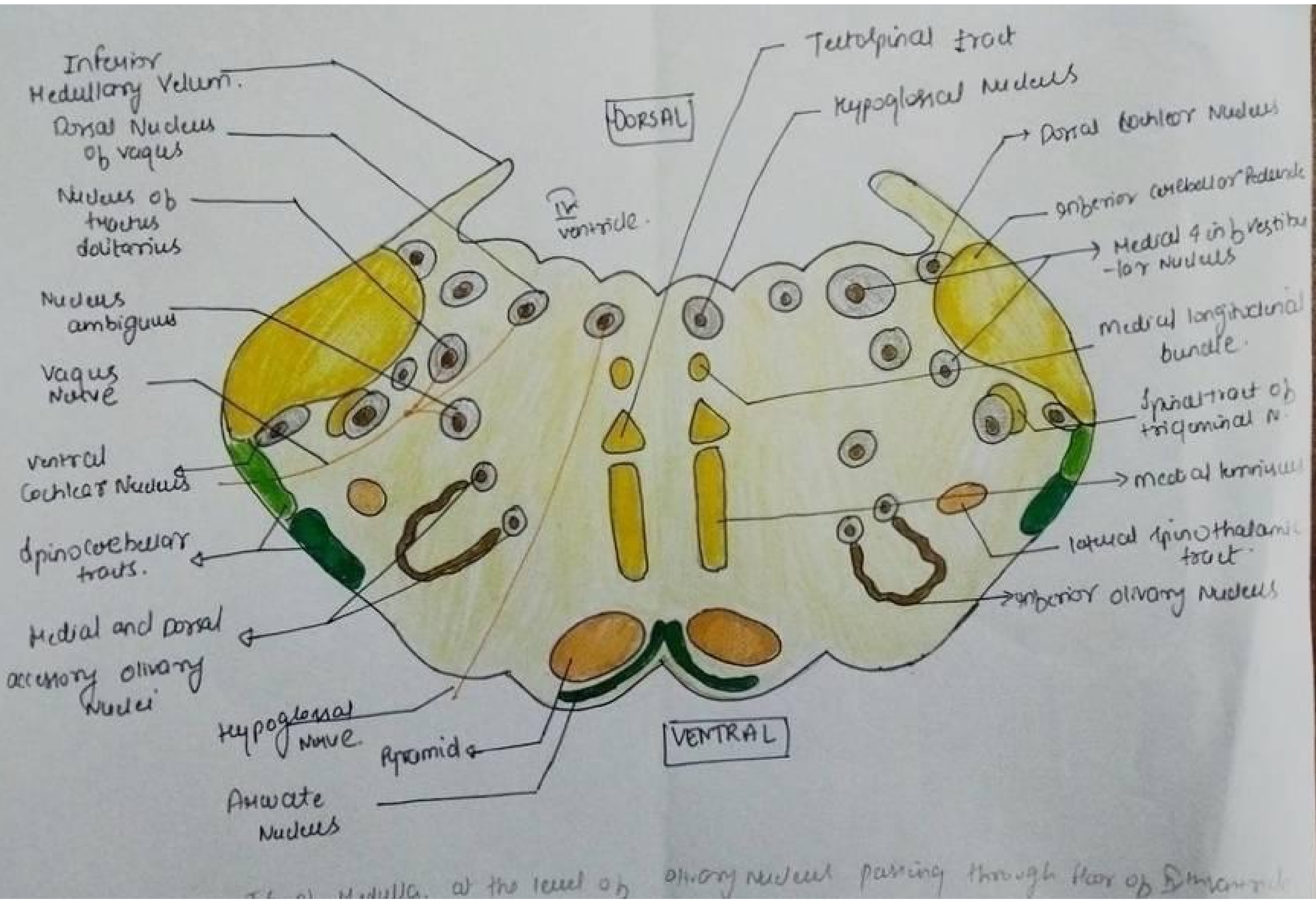
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olivary Nucleus passing through floor of

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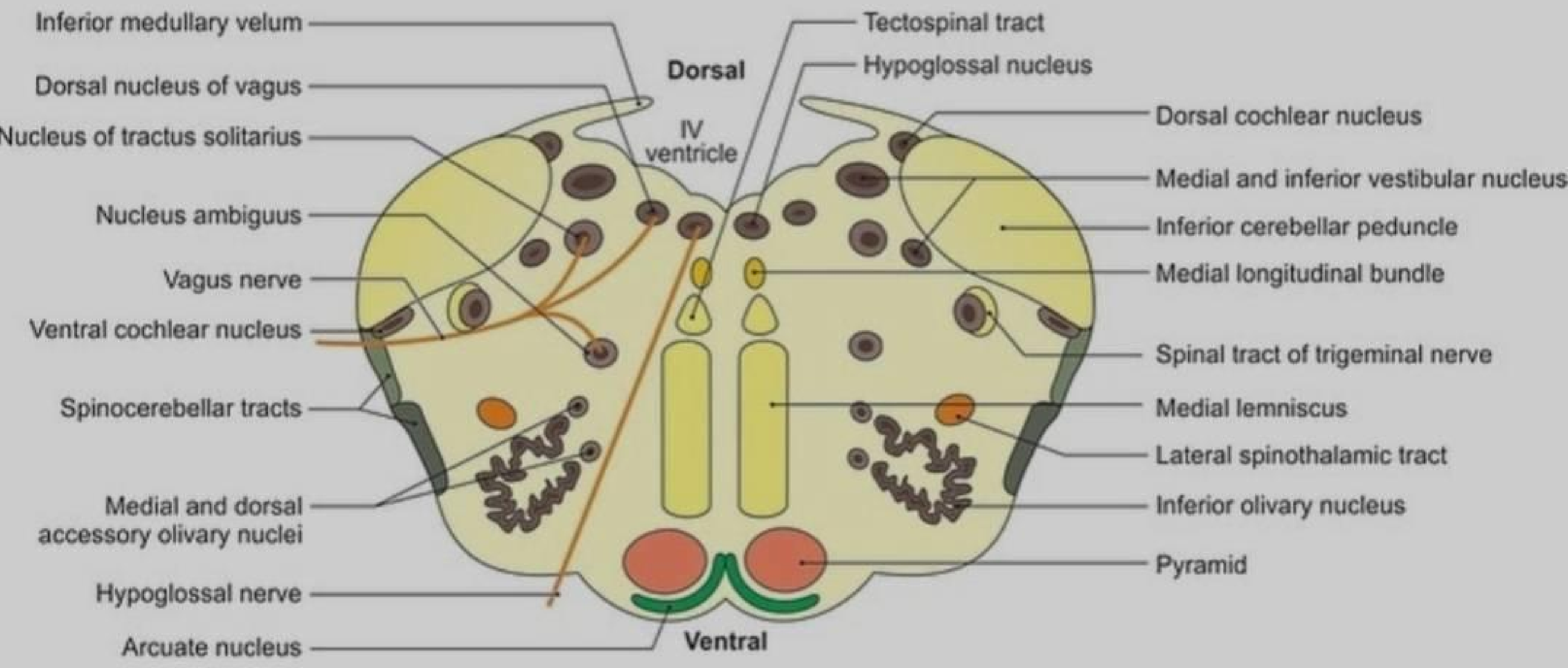


Fig. 5.7: TS of medulla oblongata at the level of olivary nucleus passing through floor of fourth ventricle

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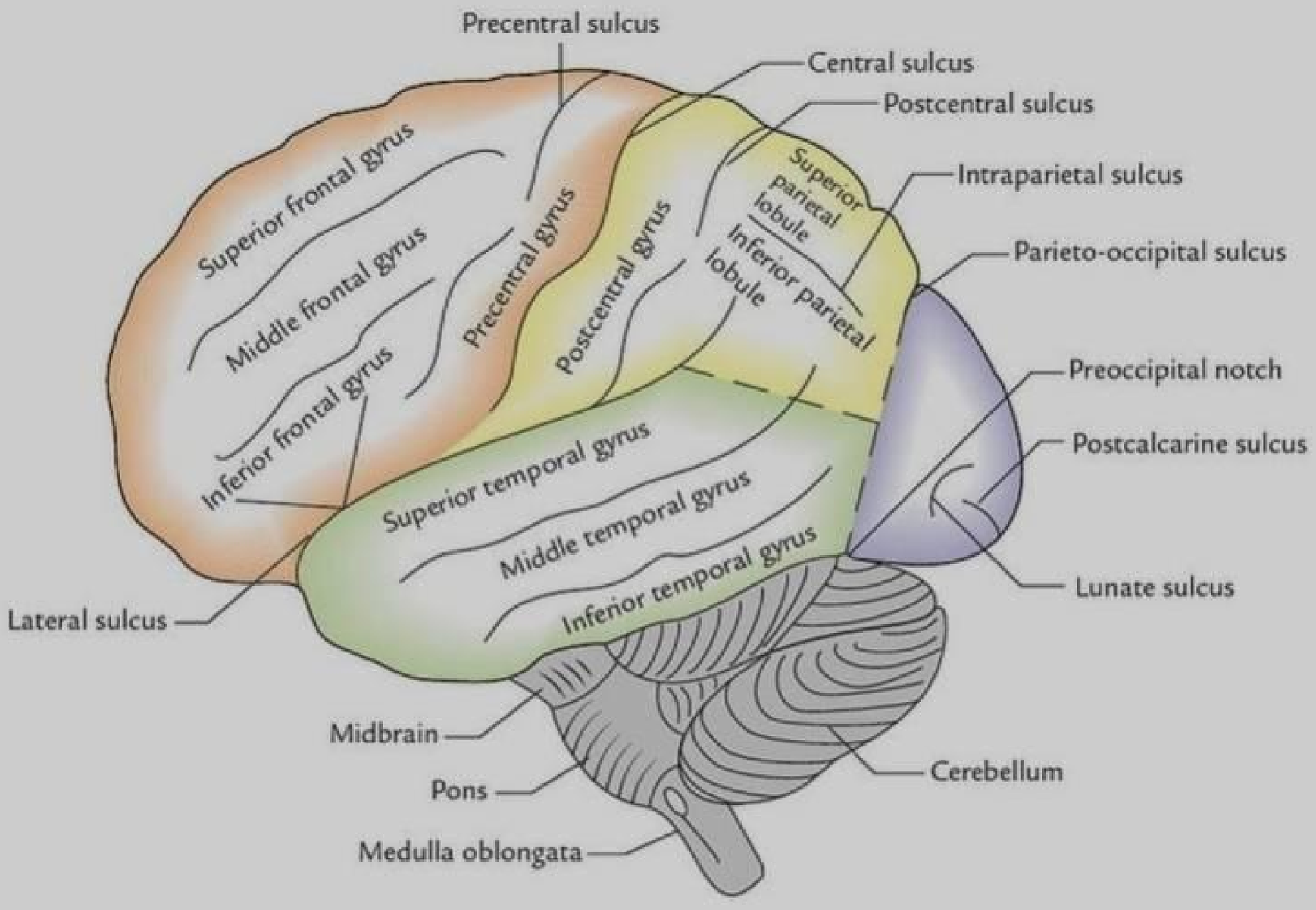


Fig. 23.2 Lateral aspect of the left side of the brain. Note the four lobes on the superolateral surface of the cerebral hemisphere.

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CEREBELLUM ...

- Cerebellum is → largest part of hind brain
- located in → posterior cranial fossa behind pons and medulla
- Each hemisphere of cerebellum → connected to brainstem by → 3 pair of large fibre tract ↓ called cerebellar Peduncles.
- function :—
  - Maintenace of equilibrium
  - Regulation of muscle tone
  - Coordination of somatic motor activities

External features :

- Parts - 3x part - 2x lateral hemispherical lobes
  - 2x cerebellar hemisphere
  - narrow median worn like portion - Vermis
- Surface - superior surface
  inferior
- Notches - Anterior aspect of cerebellum marked by → wide shallow anterior cerebellar notch
  - Posterior cerebellar notch is → deep and narrow
- Fissure - Horizontal fissure → along lateral and posterior margin of cerebellum
  - marks junction b/w - superior and inferior surface of cerebellum.
- Posterolateral surface → inferior surface of cerebellum
  - separate → flocculonodular lobe from rest of cerebellum
- V-shaped fissura prima → divide cerebellum into anterior and posterior lobes (middle)

Sub-division of Cerebellum :

Anatomical sub-divisions :- into 3x lobes - Anterior
- Posterior
- flocculonodular

- Anterior lobe :- lie on superior surface anterior to - fissure prima
- Posterior/middle lobe :- between fissura prima and posterolateral
- Flocculonodular lobe :- smallest of all.
  - lies on inferior suface

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Morphological Sub-divisions :-

- Based on phylogenetic and functional criteria

- Archicerebellum (vestibular cerebellum)
  - oldest part of cerebellum
  - consist of flocculonodular and lingula
  - function- maintainance of equilibriam, tone

- Paleocerebellum (spinal cerebellum)
  - Phylogenetically, it is next part of cerebellum to appear
  - consist of anterior lobe (except lingula) and pyramid and uvula of inferior vermis
  - function - concerned with tone, postrure and crude mruement of limb

- Neocerebellum (cerebral cereballum)
  - Phylogenetically - most recent part of cerebellum to develop
  - made up of - middle lobe, large part of cerebellum (except - pyramid and uvula of inferior vermis)

- chiefly - corticopontocerebellar in connection
  - concerned with smooth performance of skilled voluntary movement

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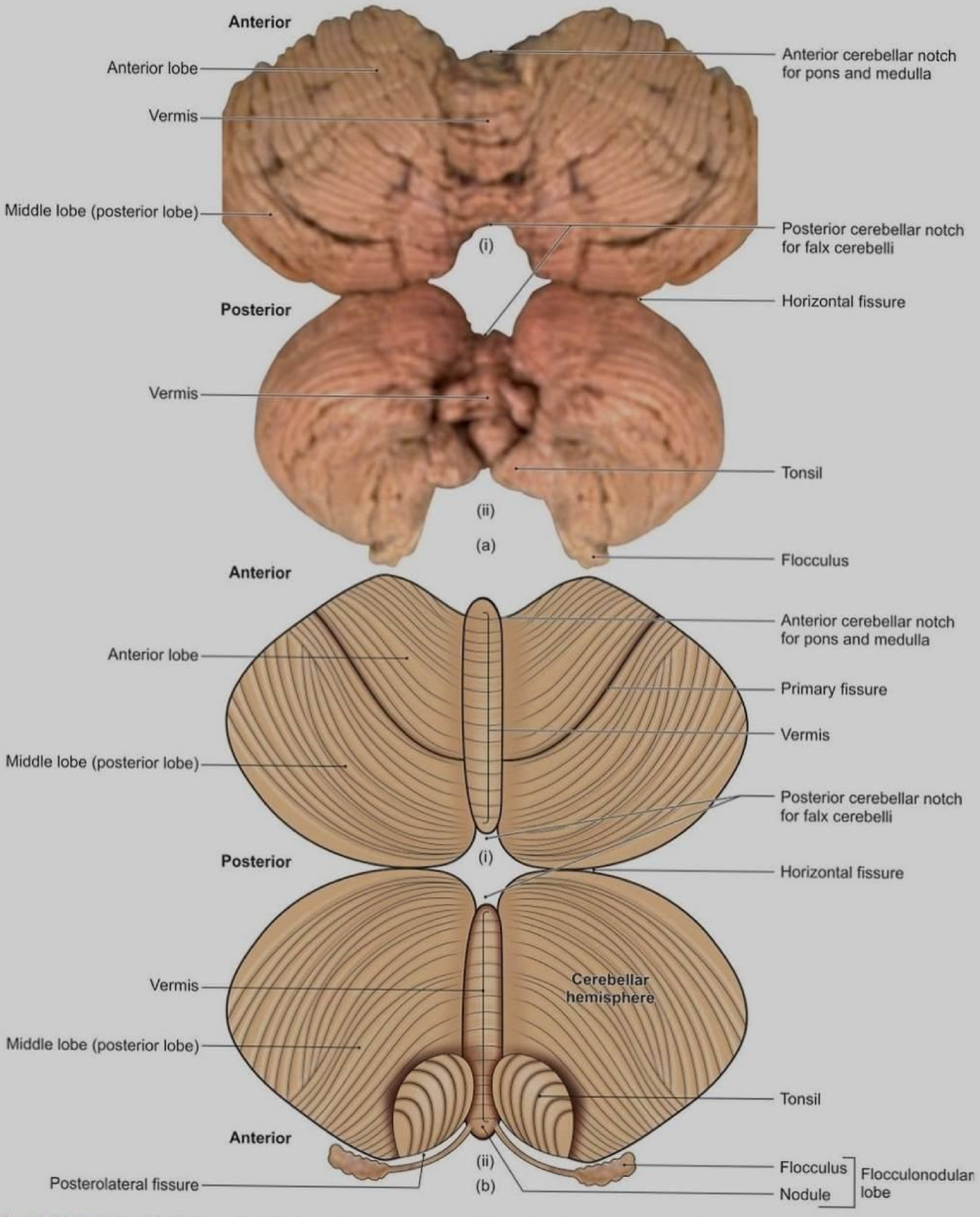
Intracerebellar Nuclei :-

- masses of grey matter embedded in the white matter of cerebellum

- Dentate nucleus :- Prominent of intra-cerebellar nuclei
  - nucleus of neocerebellum
- emboliform nucleus :- oval in shape
  - nucleus of paleocerebellum
- Globose nucleus :- rounded in shape
  - lie between emboliform and fastigial nuclei
  - nucleus of paleocerebellum
- fastigial nucleus :- lie near the midline in vermis
  - close to roof of 4th ventricle
  - nucleus of Archicerebellum

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Figs 6.3a and b: (a) (i) Superior, (ii) inferior surface of cerebellum (gross); (b) (i) Superior, (ii) inferior surface of cerebellum (diagrammatic)

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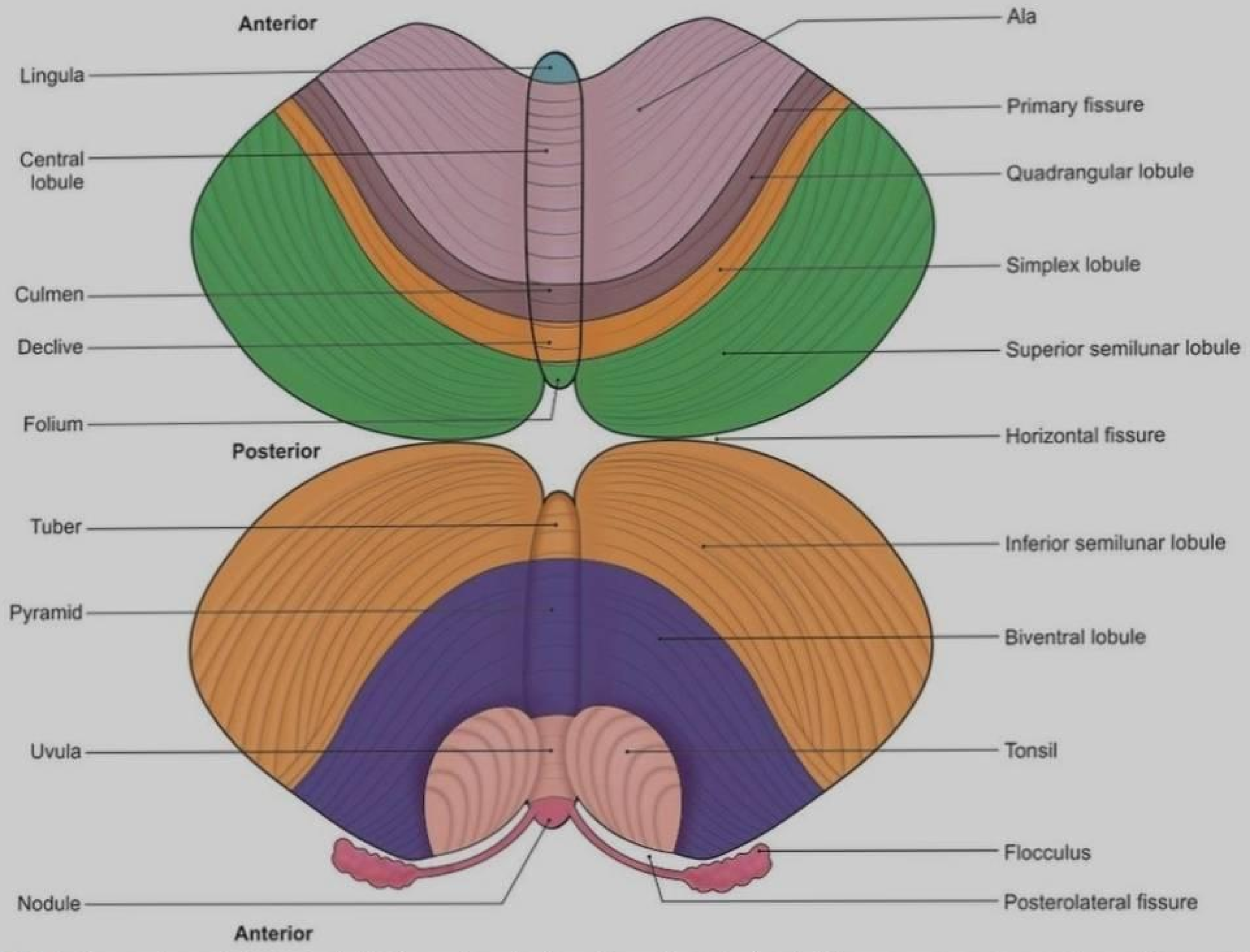


Fig. 6.4: Lobes and morphological subdivisions of cerebellum. Area above horizontal fissure represents superior surface and area below the fissure shows inferior surface

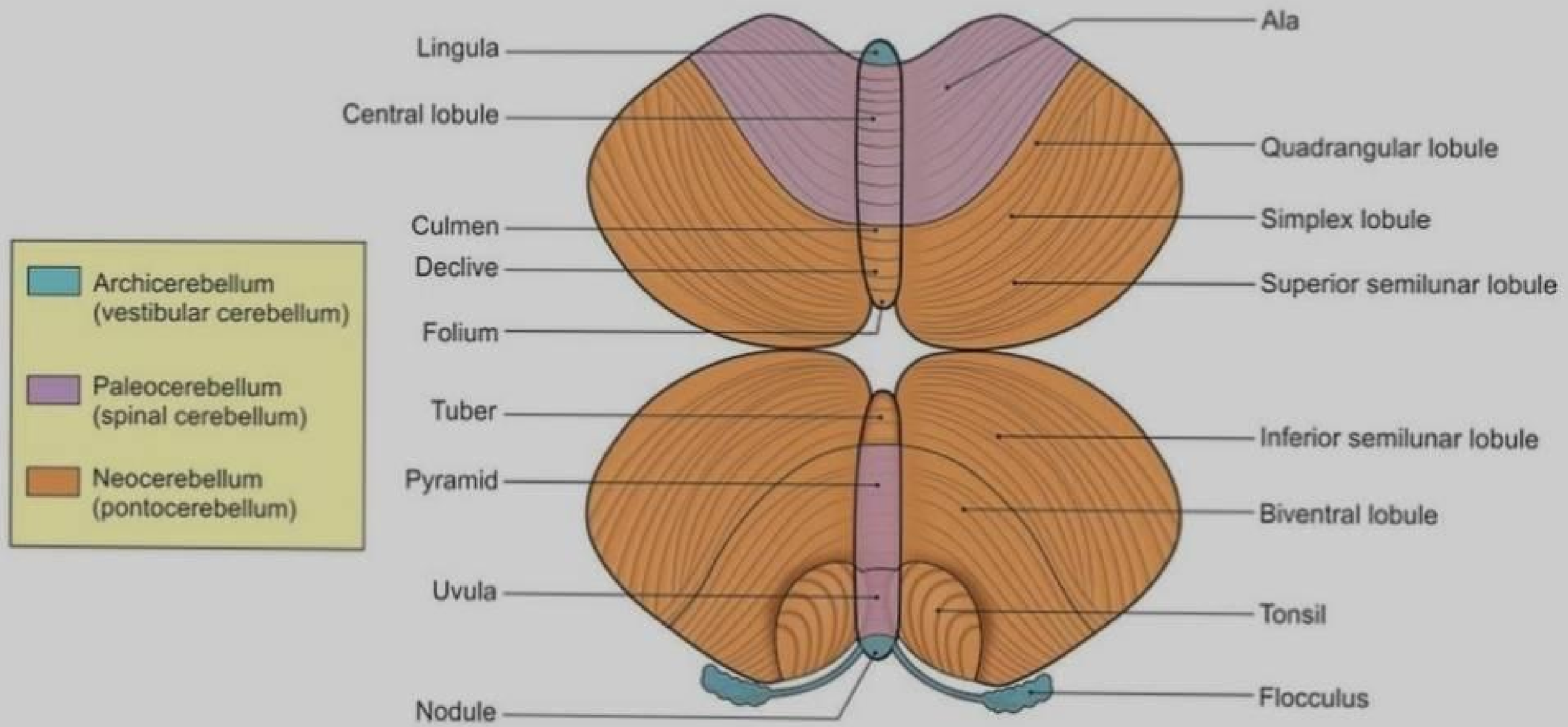


Fig. 6.5: Functional subdivisions of cerebellum

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INTERNAL STRUCTURE OF CEREBELLUM...

- Cerebellum is made up of thin surface of layer of grey matter called - cerebellar cortex
- and central core of white matter
- embedded within central core of white matter are masses of grey matter called - "intracerebellar nuclei".

Structure of Cerebellar Cortex :-

- Structure of cerebellar cortex is uniform throughout (homotypical)
- Consist of 3x distinct layer :-
  (1) Outer molecular layer
  (2) Intermediate Purkinje cell layer
  (3) Inner granular layer

- **Molecular layer** :- - mainly consists of → dendrit of Purkinje cell and few nerve cell
  - nerve cell are - basket cell and stellate cells SB I

- **Intermediate Purkinje cell layer** :-
  - consist of → single row of large flasked shaped cell - Purkinje cell.

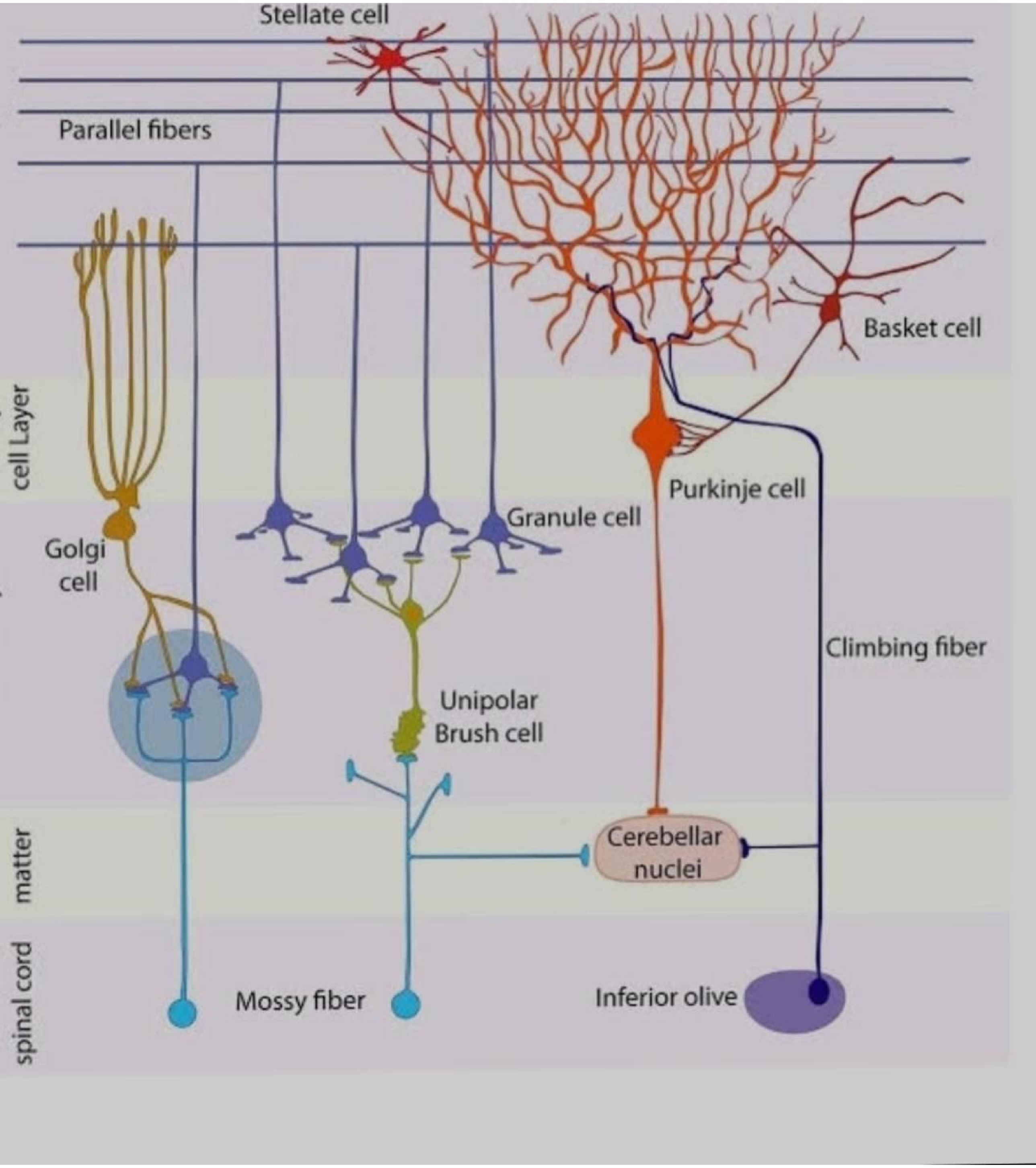
- **Granular layer** :- consist of closely packed - small granule cell
  - also contains - few large Golgi cells.

- **Intrinsic neurons of Cerebellar Cortex** :-
  - 5x type of intrinsic neurons in cerebellar cortex
    - Purkinje Cell
    - Granule cell
    - Stallate cell
    - Basket cell
    - Golgi Cell

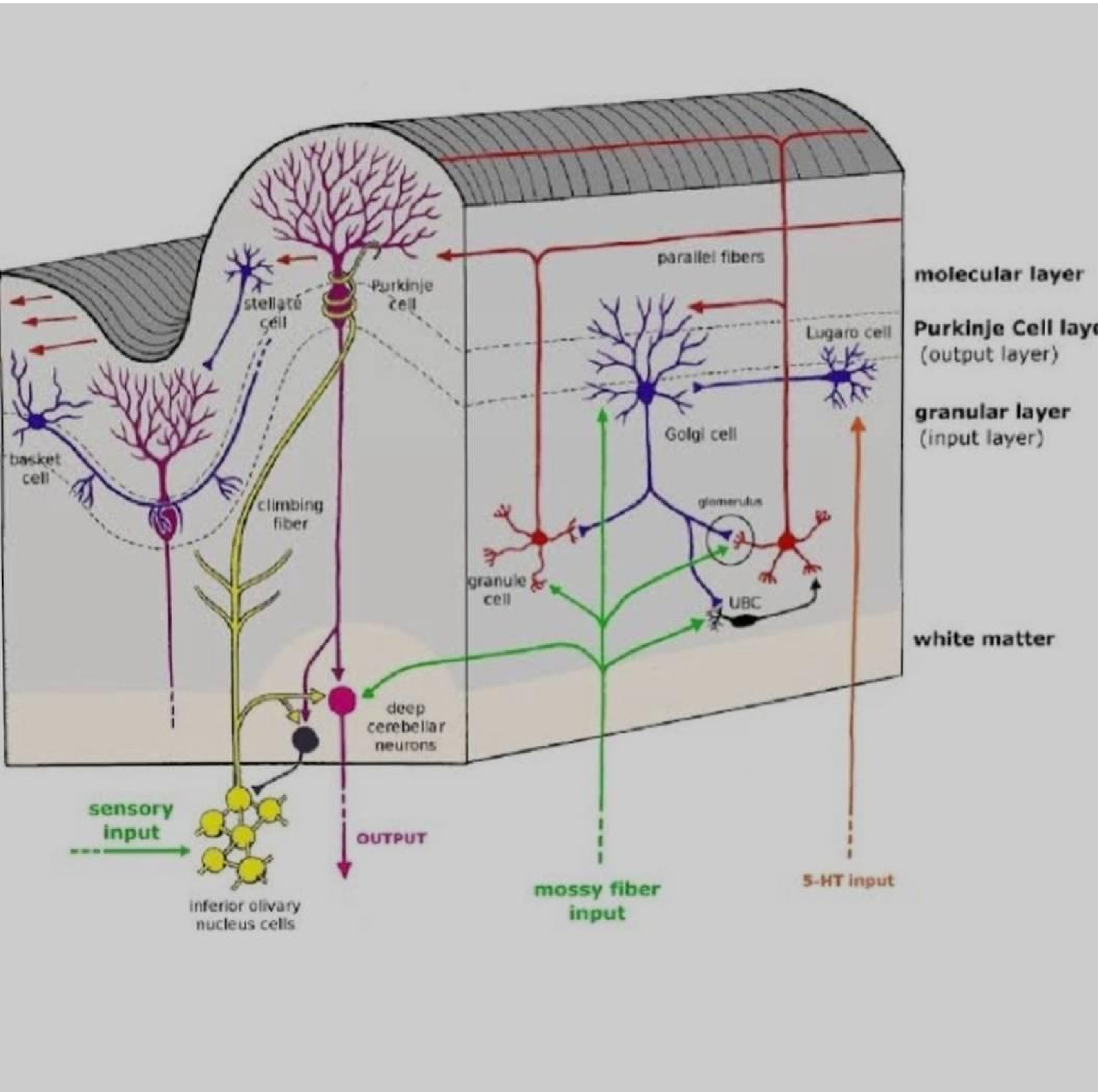
- **Intra-Cerebellar nuclei** :-
  - Masses of grey matter embedded in white matter of cerebellum
  - On each side of → midline
  - 4x in number

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THALAMUS ...

- Thalamus → large mass of grey matter
- 4x cm in each - transverse
  vertical
  antero-posterior diameter
- situated in → lateral wall of 3rd ventricle
  and
  floor of central part of → lateral ventricle

Structure and nuclei of thalamus :-

- White matter → external medullary lamina cover → lateral surface
  - internal medullary lamina divide → thalamus in → 3x parts
    - Anterior
    - Medial
    - lateral

- Grey matter - Divided into 7x nuclei
  ↓
  Table

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Thalamus

Anterior nuclei of thalamus

Lateral dorsal nucleus

Thalamic reticular nucleus

Ventral anterior nucleus

Ventral lateral nucleus

Lateral posterior nucleus

Ventral posteromedial nucleus

Ventral posterolateral nucleus

Mediodorsal nucleus

Lateral nuclei of thalamus

Periventricular nuclei of thalamus

Medial medullary lamina

Intralaminar nuclei of thalamus

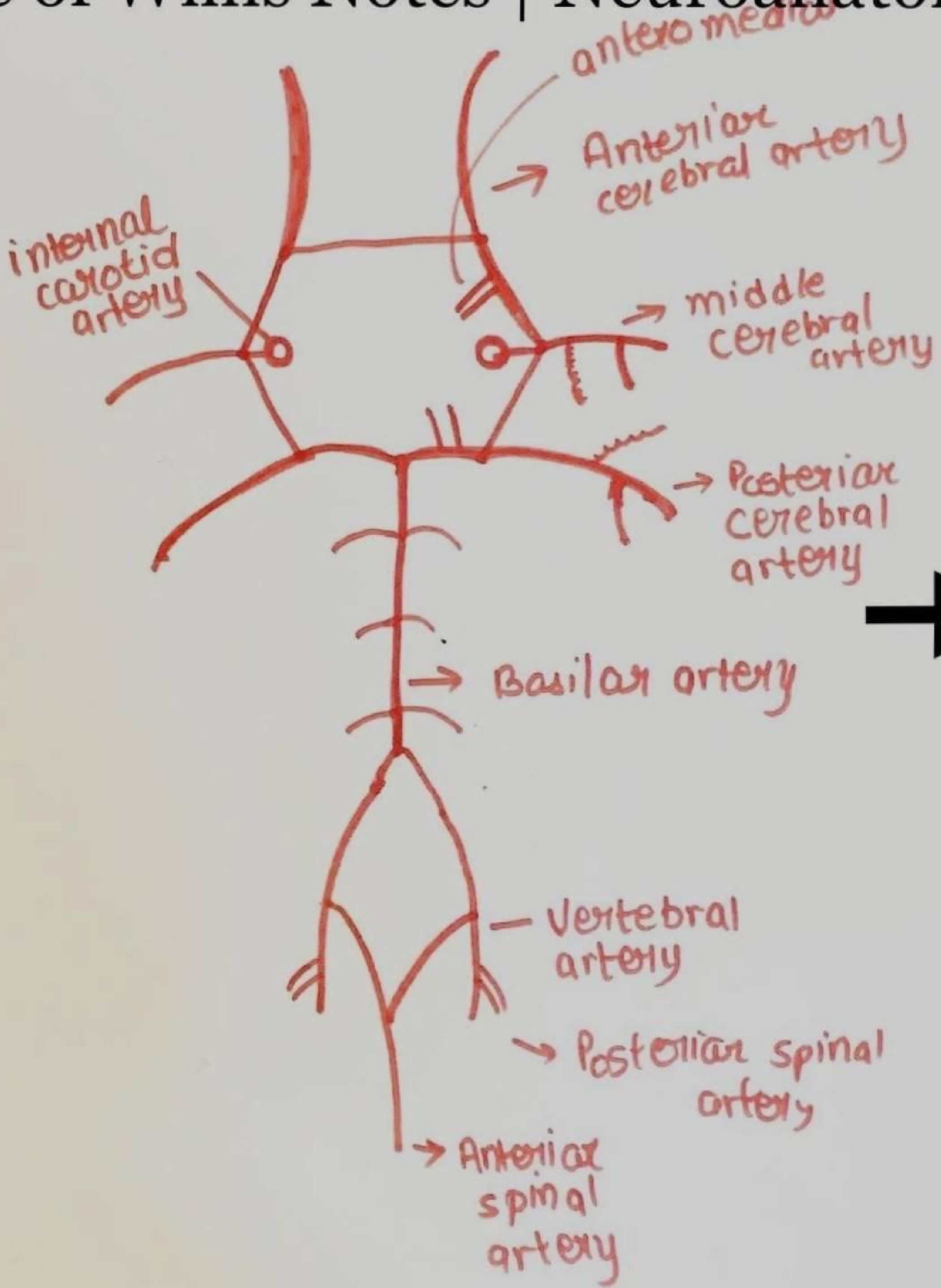
Pulvinar of thalamus

Medial geniculate body

Lateral geniculate body

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Circle of Willis Notes | Neuroanatomy



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Circulus Arteriosus / Circle of Willis...

Hexagonal arterial circle
situated at → Base of Brain in → interpeduncular fossa

- formed by ◦- Anterior cerebral branch, internal carotid
◦- terminal part of internal carotid arteries
◦- Posterior cerebral branch, basilar artery

◦- Two anterior cerebral arteries connected by → Anterior communicating artery

◦- internal carotid and posterior cerebral artery of same side ↓ united by Posterior communicating artery

Formation :-

Anteriorly- Anterior communicating artery
Anterolaterally - Anterior cerebral artery
laterally - internal carotid arteries
Posterolaterally - Posterior communicating artery
Posteriorly - Posterior cerebral arteries

Branches :-

◦- Cortical branches supply → cerebrum (anastomose freely)
◦- central branches → thalamus, corpus striatum, internal capsule (does not anastomose)
◦- Choroidal branch → choridal plexuses of ventricle

1.] Cortical branches are → Anterior, Middle, Posterior cerebral arteries

◦- Supply - cerebrum → Superiolateral Surface mainly by → middle cerebral artery
◦- except → occipital lobe - PCA
- temporal lobe → PCA
- 2cm wide superior medial border ↓ frontal Pole to Parieto-occipital Sulcus → ACA

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→ medial surface mainly by → Anterial cerebral artery

- Except → occipital lobe, temporal lobe → PCA

→ inferiar surface by → Posteriar cerebral artery

- Except → medial 1/3 orbital lobe - ACA
lateral 2/3 temporal lobe – MCA

2.) Central branches :-

- Anteromedial arises → Anterior cerebral artery
  - enter → medial most part of → anterior Perforated substance

- Anterolateral arises → middle cerebral artery
  - supply → mediate striate → Caudate nucleus, → internal capsule
  - lateral striate enter → internal capsule

- Posteromedial arises → Posterior cerebral artery

- Posterolateral arises → Posteriar cerebral arteries

3.) Choroidal branch :-

- Anterior choroidal branch → internal carotid artery
  supplies → inferior horn of lateral ventricle

- Posterior choroidal branch → Posteriar cerebral
  supplies → Rest part of choroid plexus of lateral ventricle

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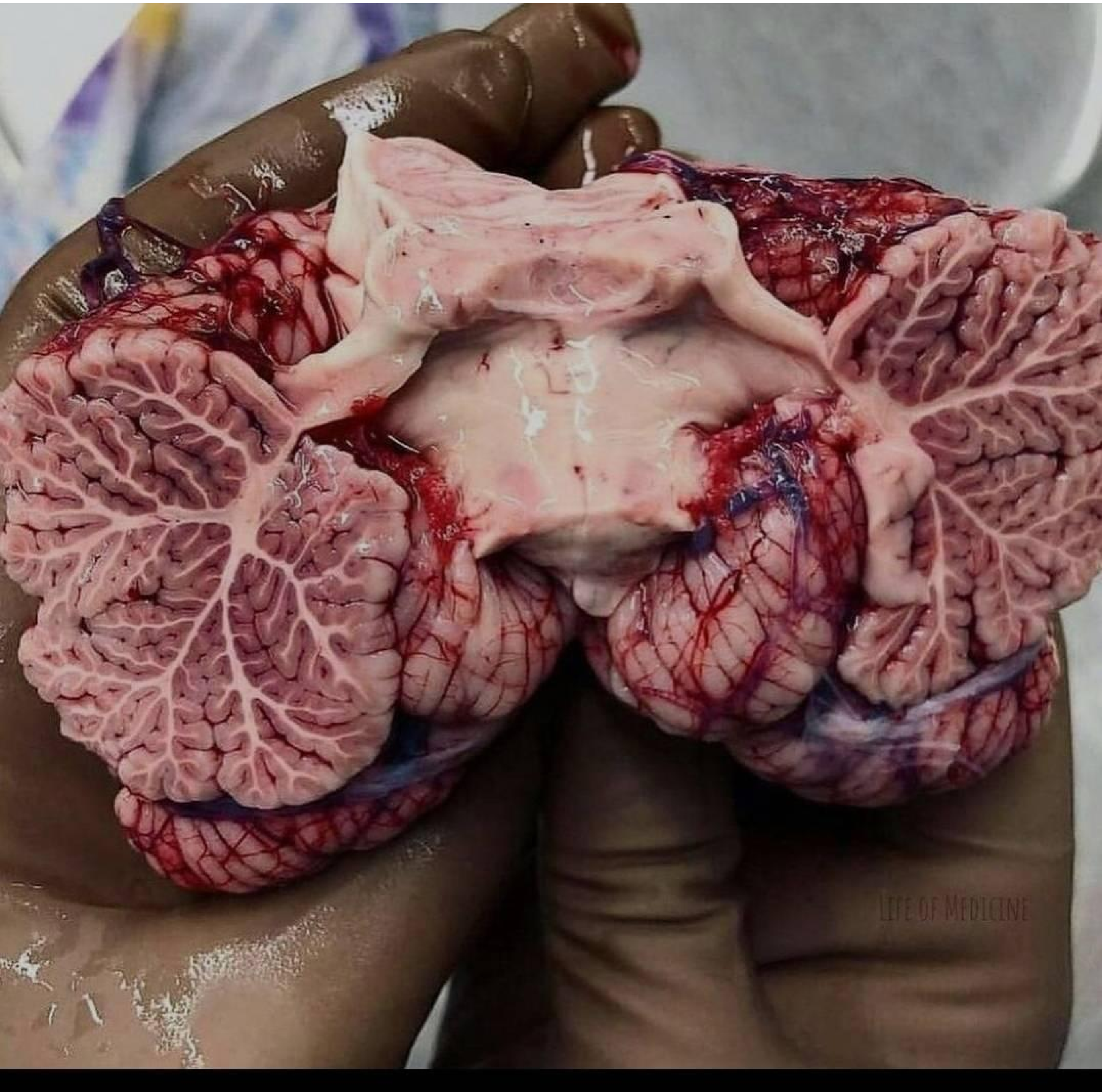
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3D ANATOMY MODELS

PREVIEW





Circle of Willis

(CIRCULUS ARTERIOSUS CEREBRI)

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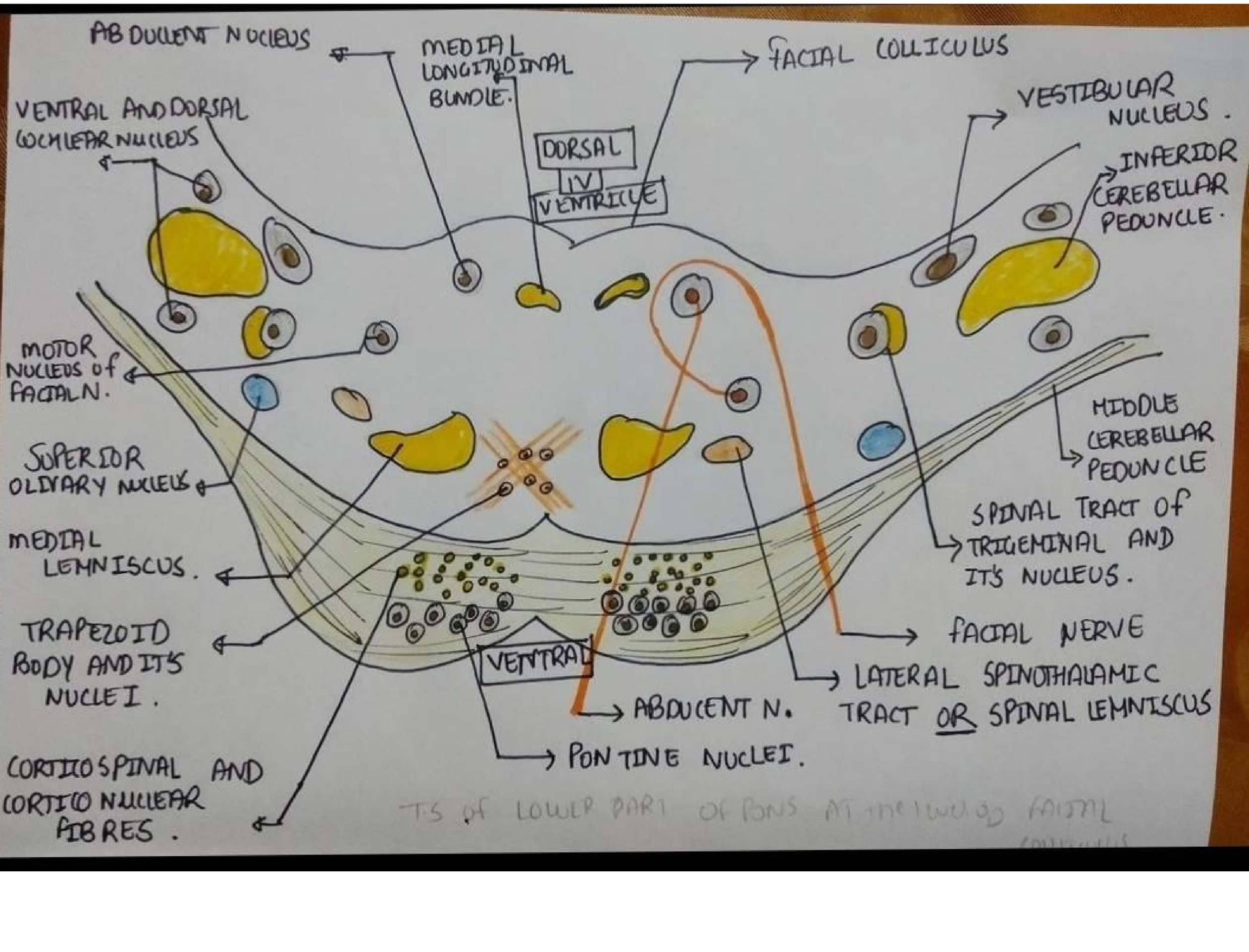
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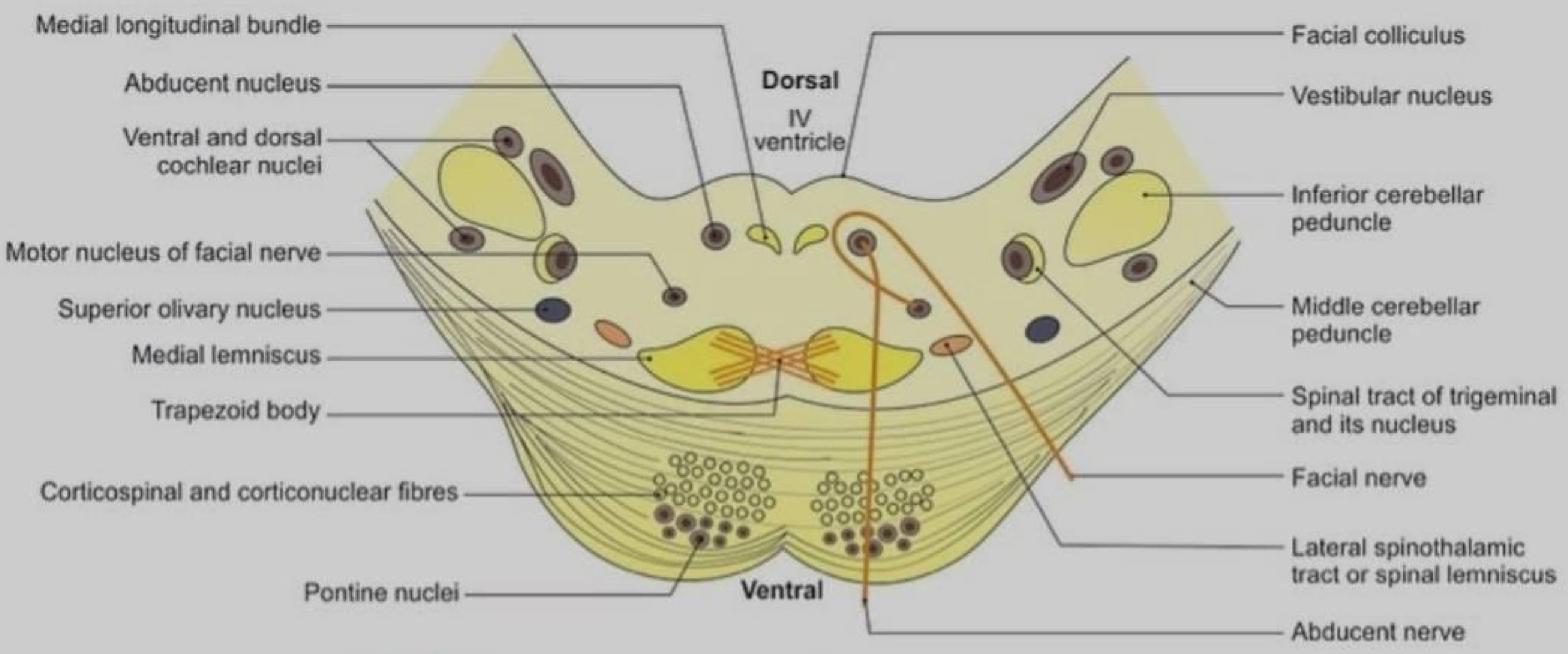
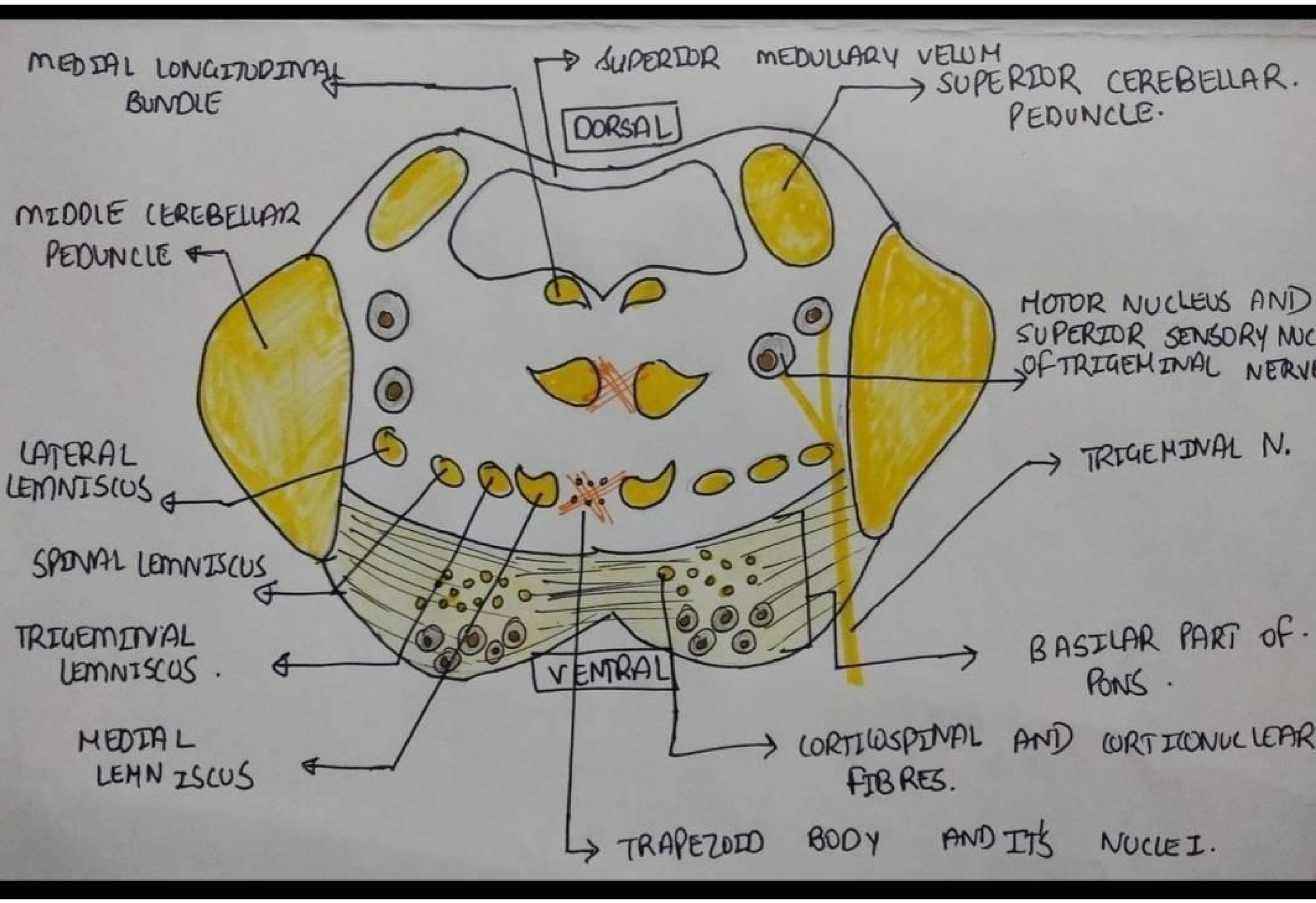


Fig. 5.9: TS of lower part of pons or TS at the level of facial colliculus

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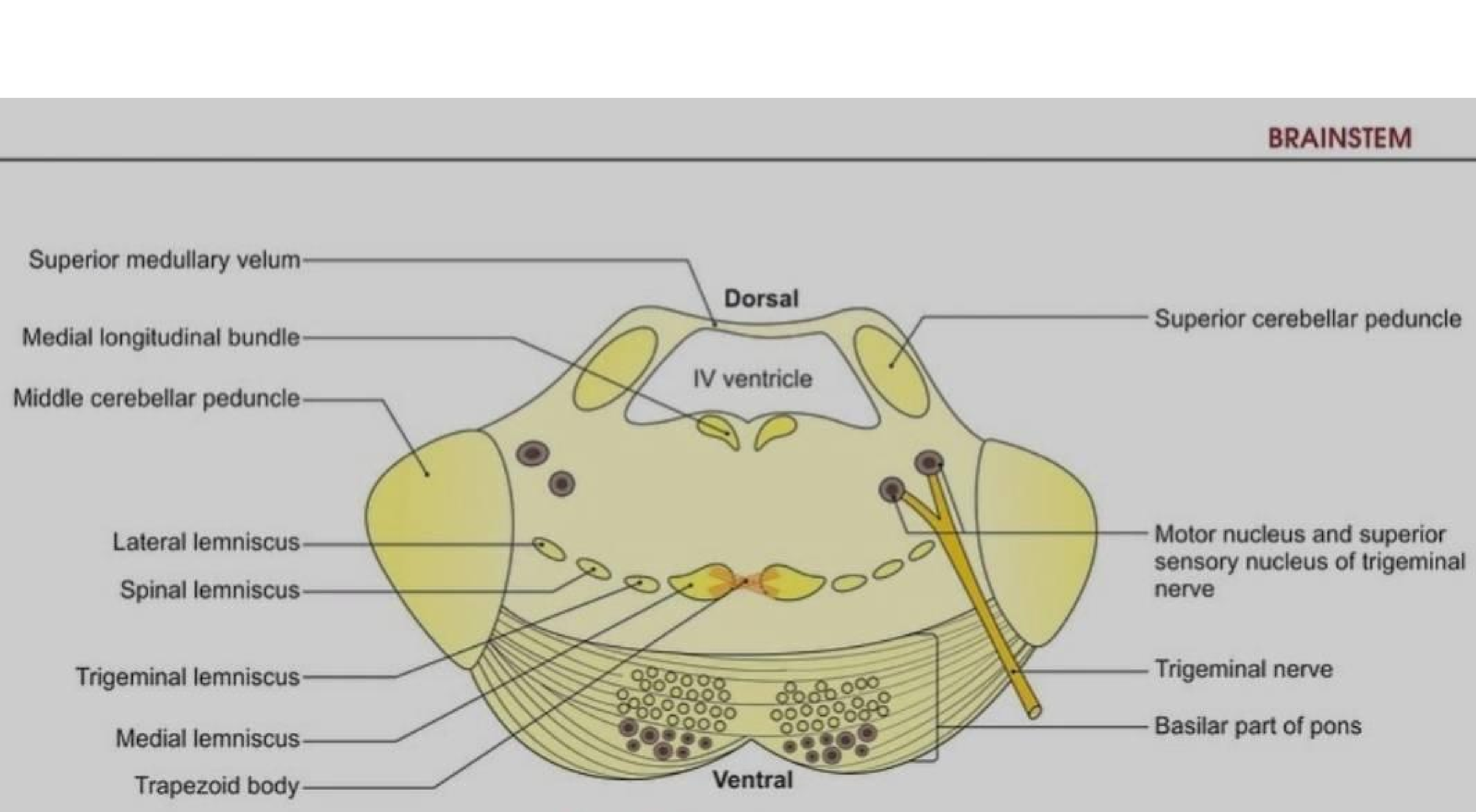
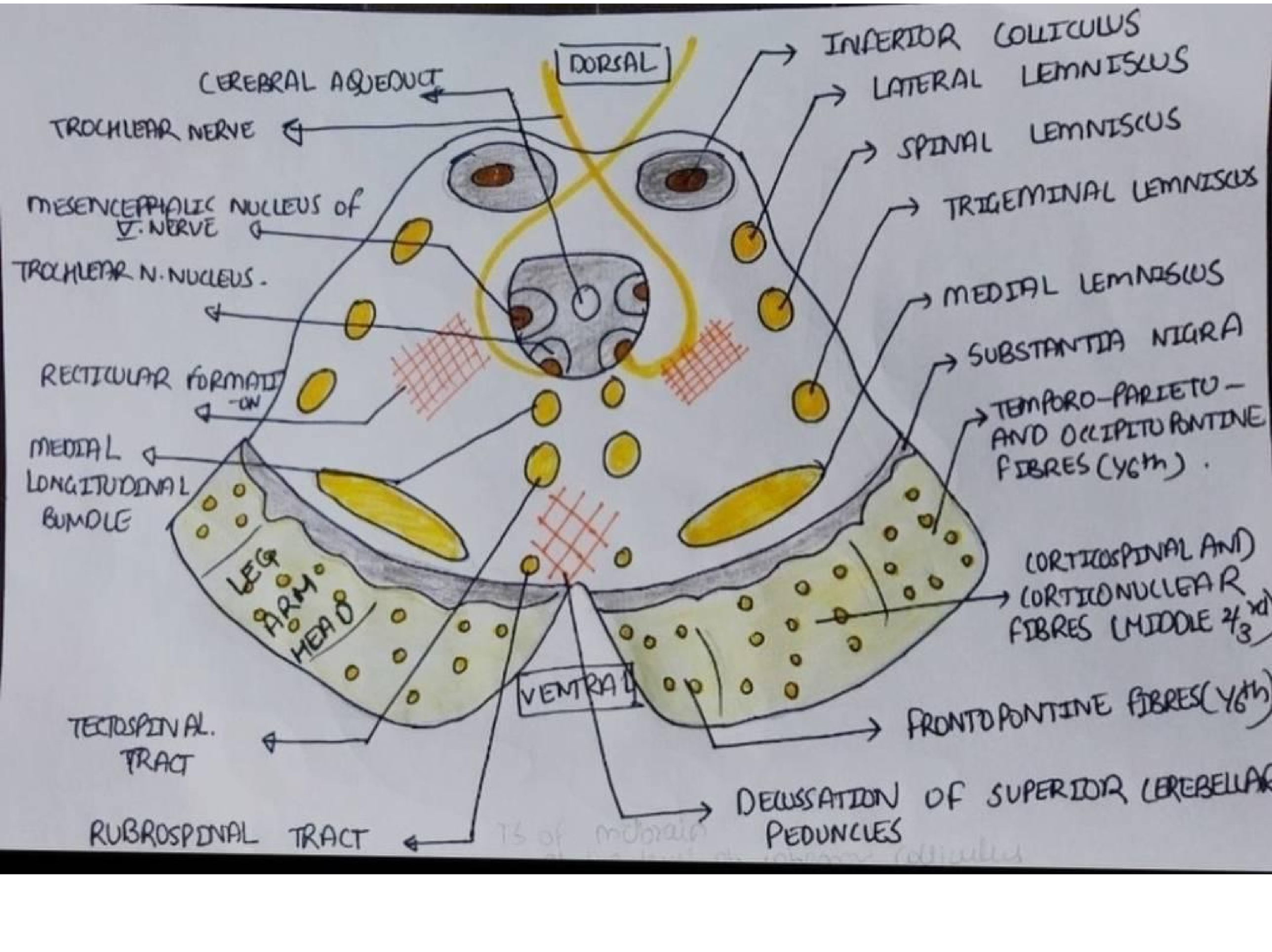
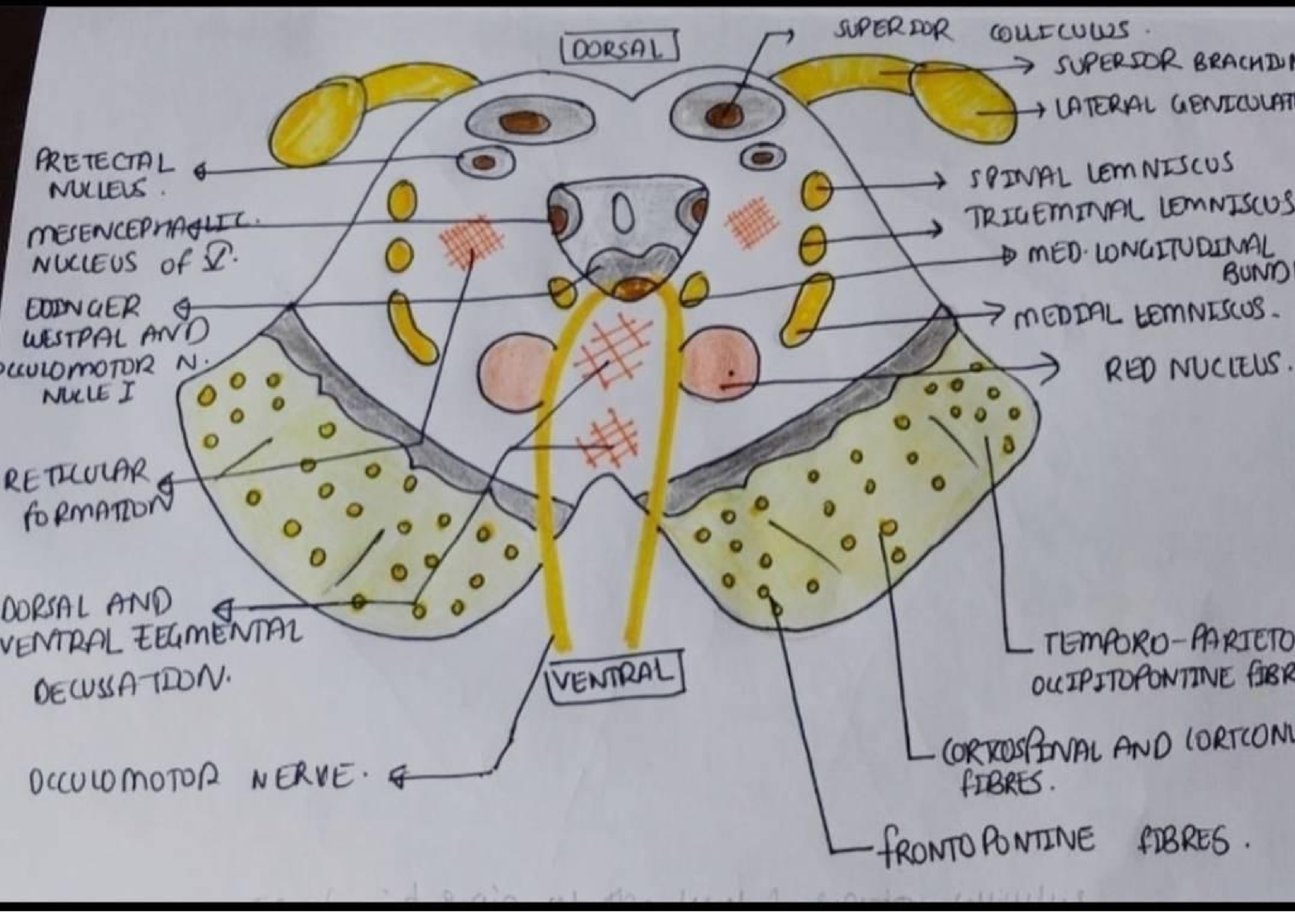


Fig. 5.10: TS of upper pons

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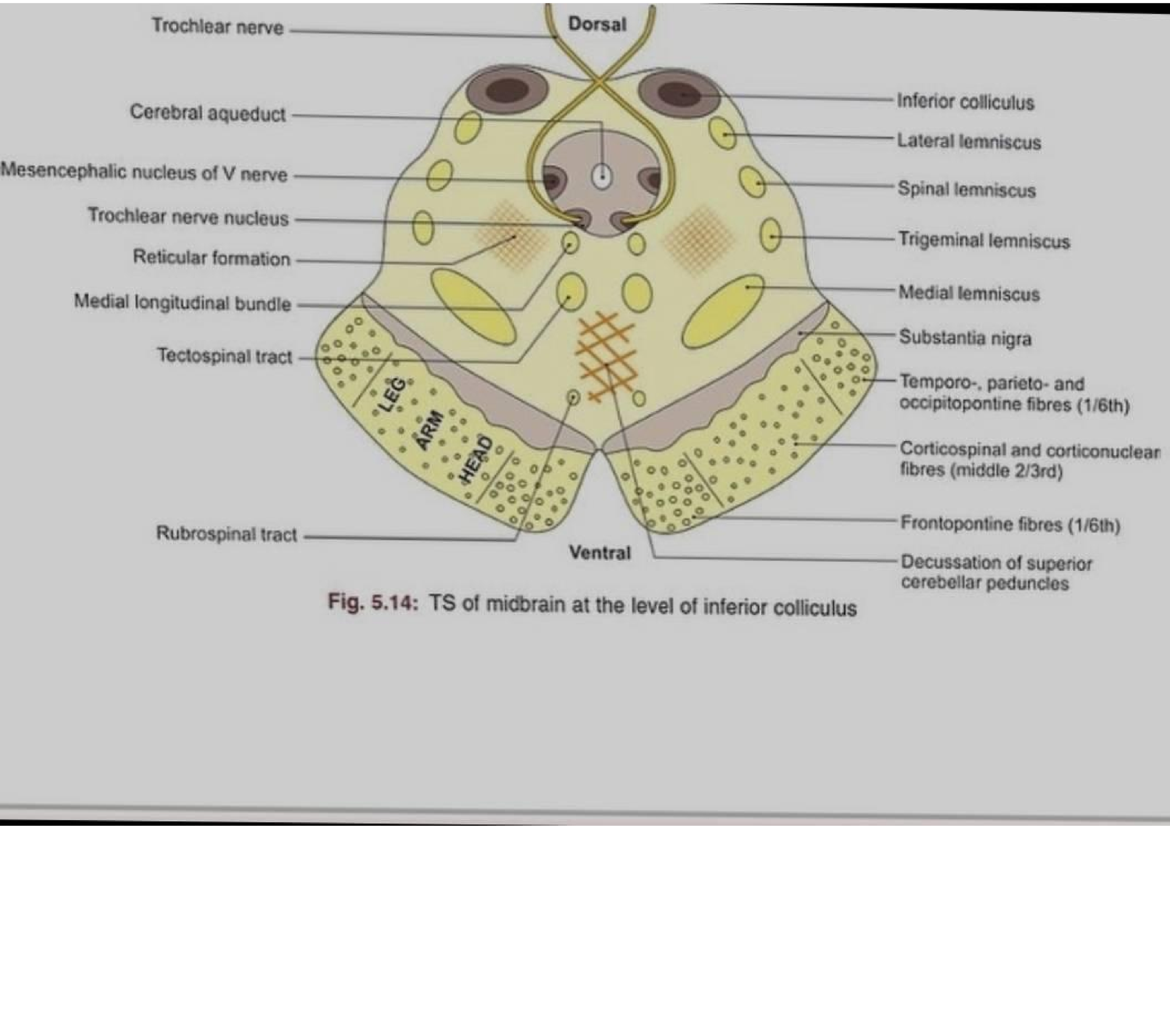


Fig. 5.14: TS of midbrain at the level of inferior colliculus

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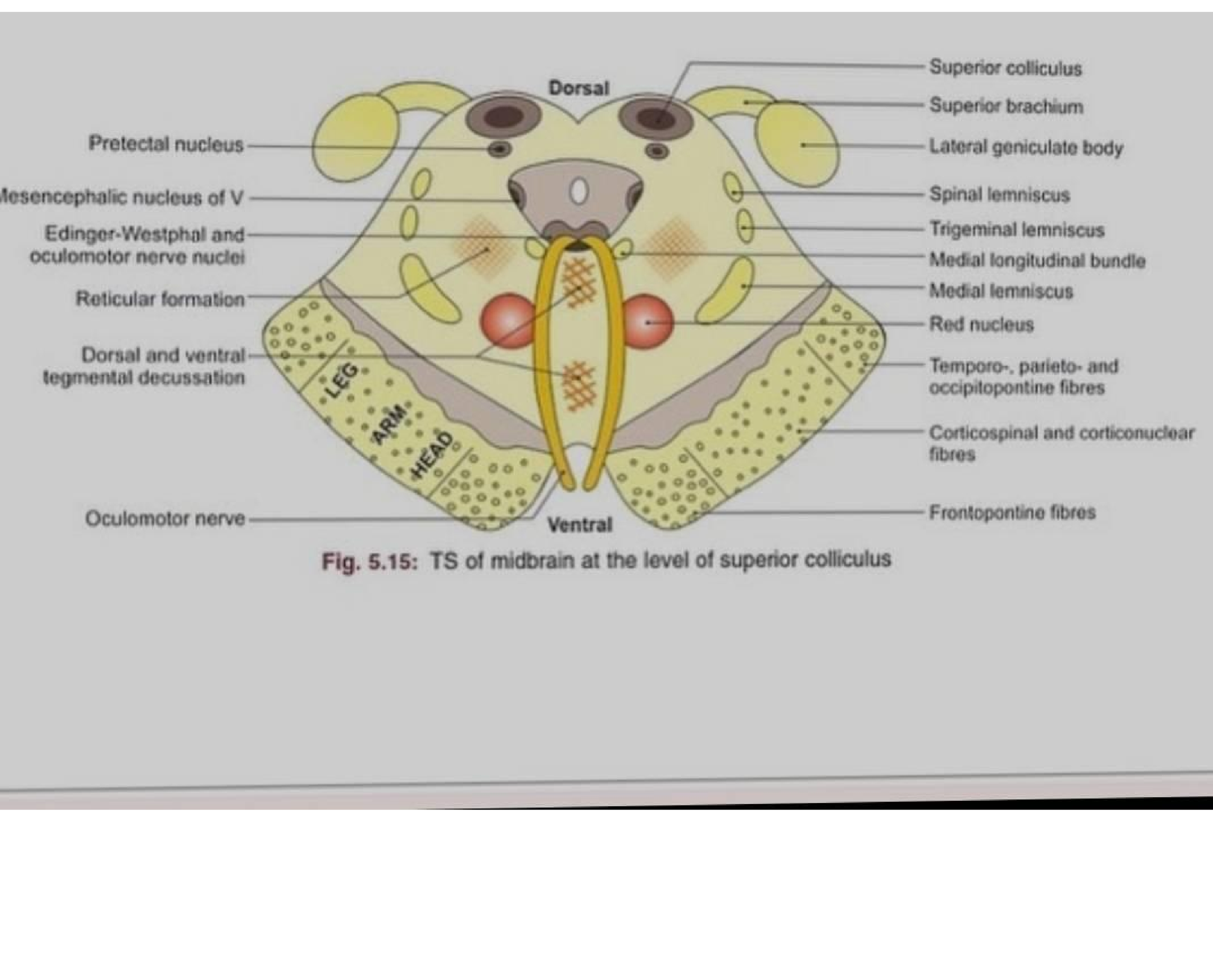


Fig. 5.15: TS of midbrain at the level of superior colliculus

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Blood Supply of spinal Card ...

* Artery :- spinal card receive → its blood supply from 3x longitudinal arterial channel
↓ extend
along length of card

:- 1x anterior spinal artery run in anterior median sulcus

:- 2x posterior spinal artery → Posterolateral sulcus

:- In addition to these channal → Pia matter has → Arterial Plexus (arteria vasocarona)

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:- main source of blood to spinal arteries is → vertebral arteries
↓ reach only up to cervical segments of card

:- spinal arteries also receive Blood from → radicular arteries
↓ reach the card
along the root of spinal nerve

:- radicular arteries arises from → spinal branch of vertebral
:- ascending cervical artery
:- Deep cervical artery
:- intercostal artery
:- lumbar and sacral arteries

:- one of radicular branches large called → arteria radicularis magna
↓
Position is variable

Vein :- draining the spinal card are arranged in form of → 6x longitudinal channels

:- unpair → 1x anteromedian, 1x posteromedian channel lie in → mid lines

:- Pair → 2x Anterolateral, 2x Posterolateral → lateral sulcus

:- these channel are → interconnected by plexus of vein (venous vasocarona)

radicular vein ← drained

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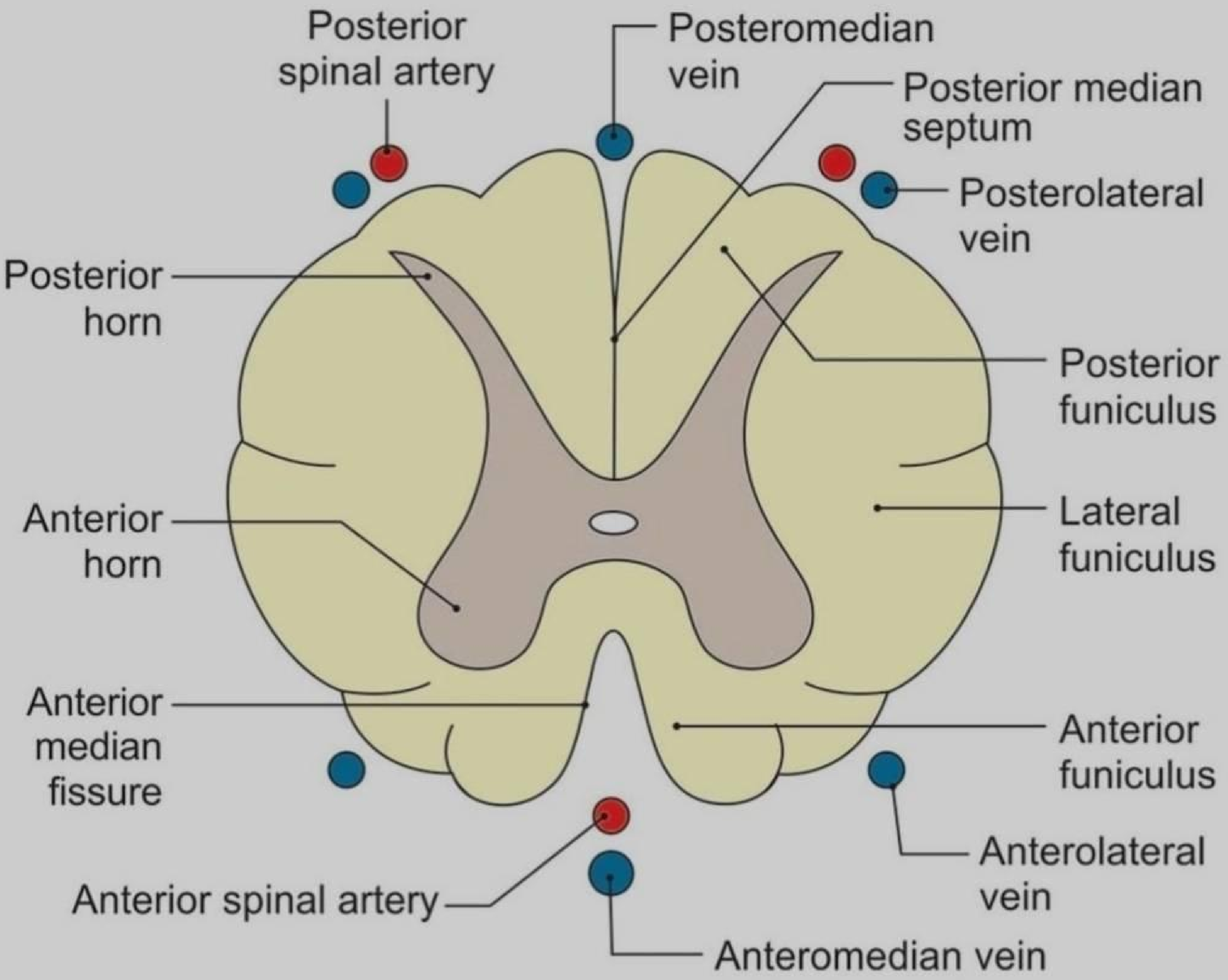


Fig. 11.1: Blood supply of spinal cord

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LATERAL VENTRICLE...

- Two irregular cavities in → cerebral hemisphere
- Communication → third ventricle through → interventricular foramen (foramen of Monro)
- Divide → central part
  → 3x horns → Anterior horn
  → Posterior horn
  → inferior horn

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Central Part:
- extend → interventricular foramen to → splenium of corpus callosum
- Boundaries:
  - Roof - Corpus callosum
  - floor - Body of caudate nucleus
    - thalamostriate vein
    - thalamus
  - Medial wall - Septum pellucidum
    - Body of fornix

Anterior horn:
- extend → front of interventricular foramen Extend into → frontal lobe
- Boundaries:
  - Anterior → Posterior surface of → genu / Rostrum of corpus callosum
  - Roof → Corpus callosum
  - floor → caudate Nucleus
  - medial → septum pellucidum / column of fornix

Posterior horn:
- lie behind splenium of corpus callosum and extend into → occipital lobe
- Boundaries:
  - floor and medial wall → forceps major
  - Roof and lateral wall → Tapetum of optic radiation

Inferior horn:
- begins at junction of central part with Posterior horn extend into → temporal lobe
- Boundaries:
  - Roof and lateral wall → Tail of Caudate nucleus
    - Amygdaloid body
  - floor → Hippocampus

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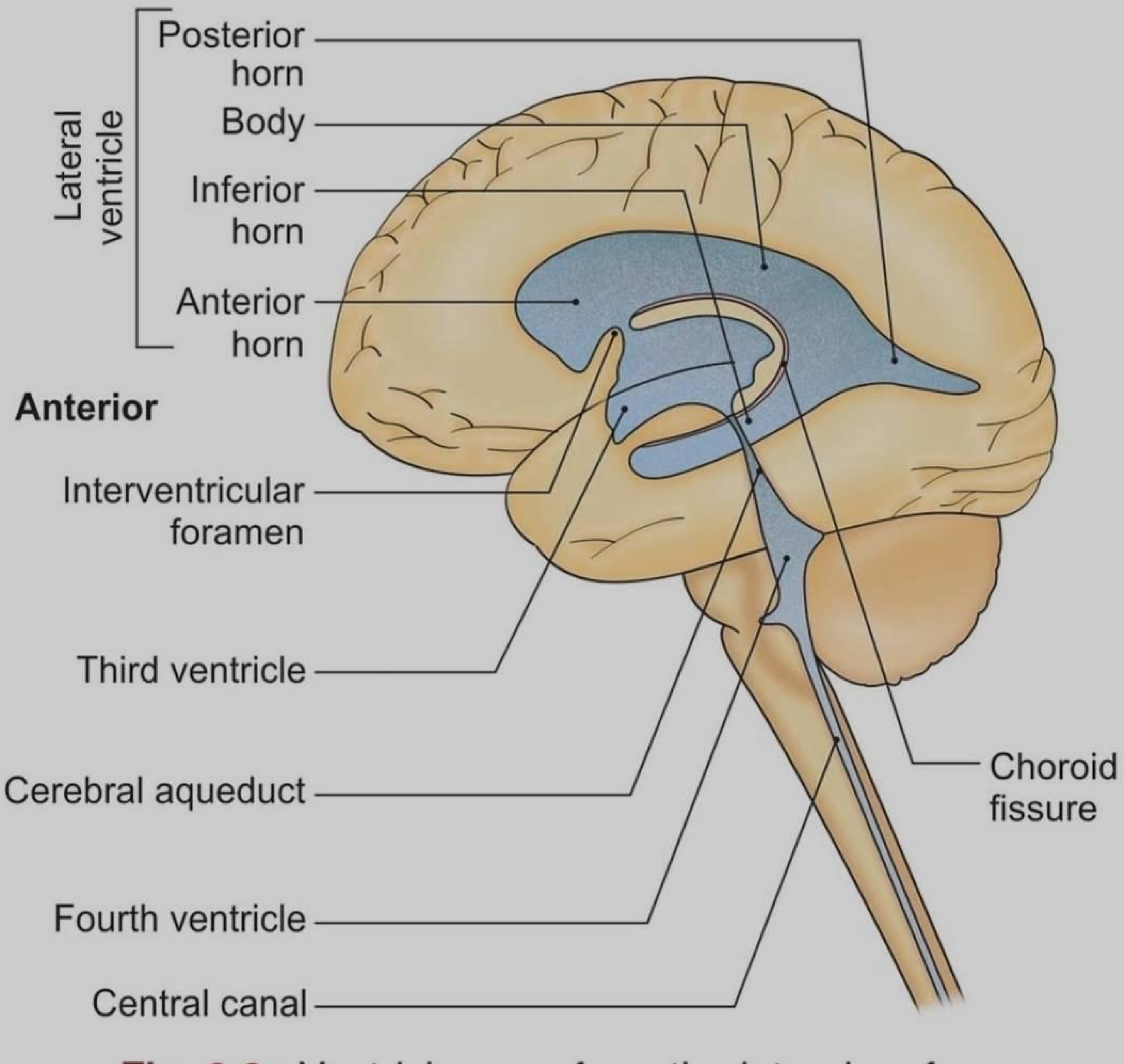
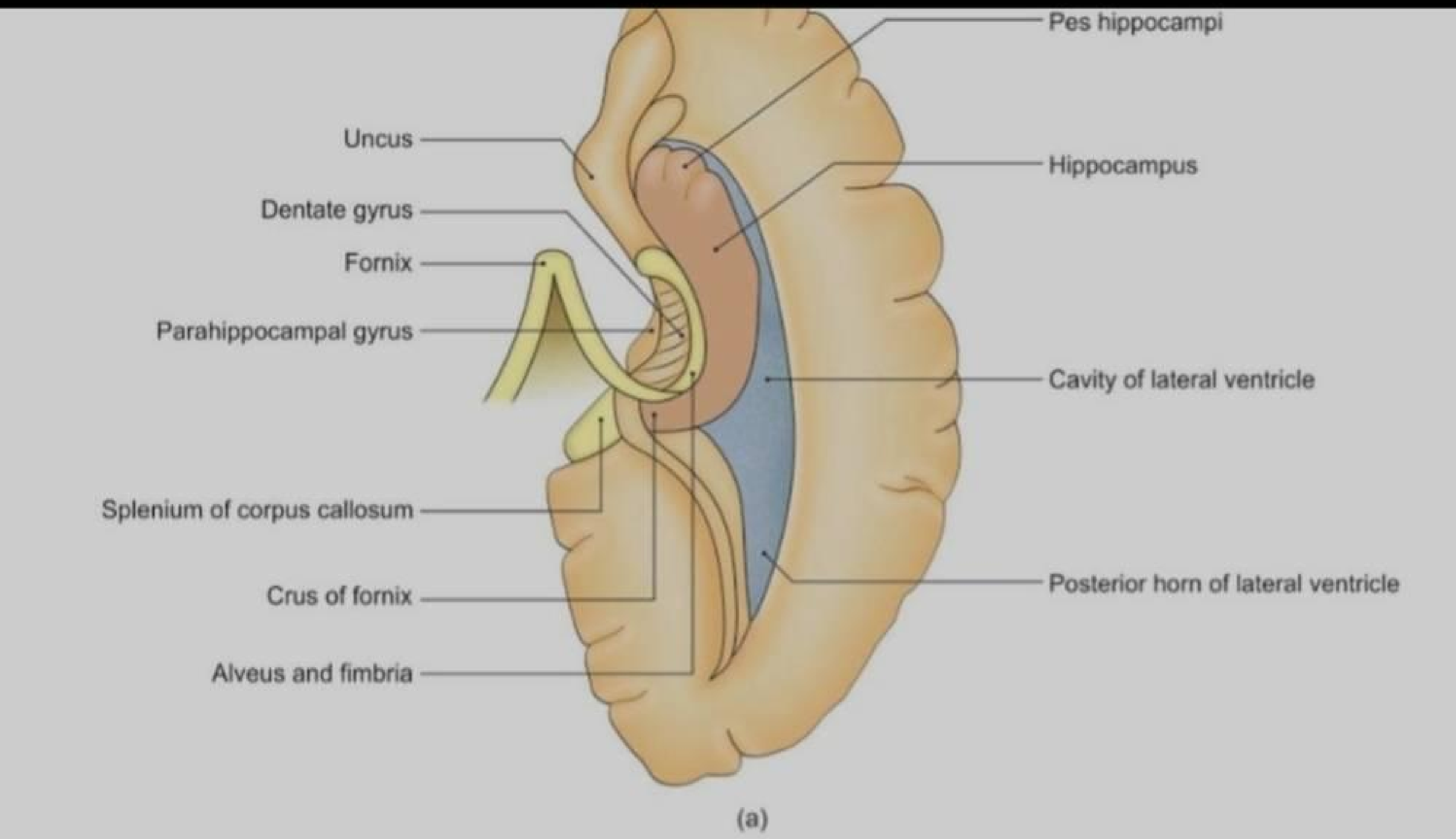
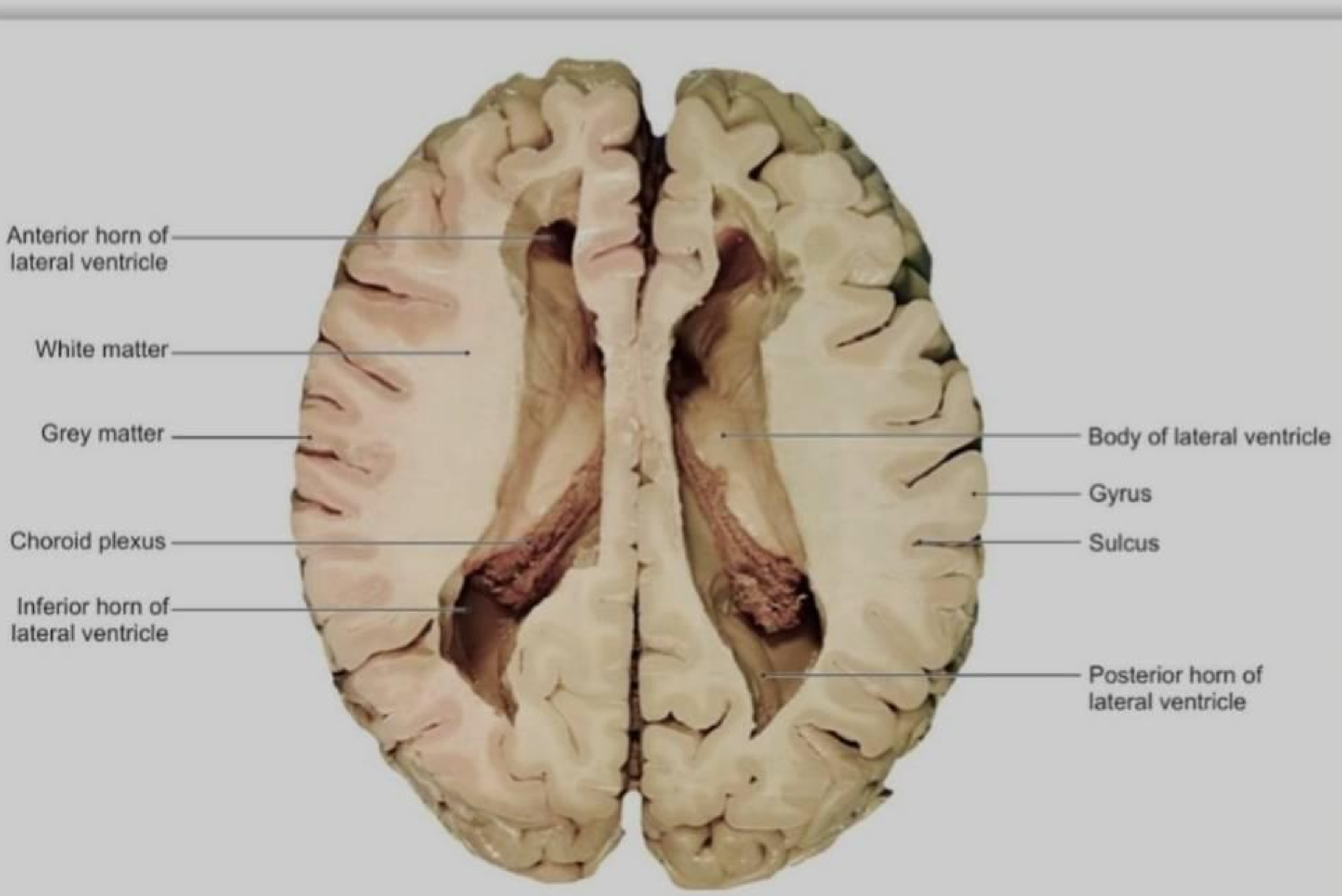


Fig. 9.6: Ventricle seen from the lateral surface

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(a)



(b)

Figs 9.10a and b: Cavity of lateral ventricle including its anterior horn, body, inferior and posterior horns

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THIRD VENTRICLE ...

- Diencephalon cavity is a → median cleft blw two thalami
- Third and lateral ventricle secrete → cerebro-spinal fluid

Communications :-

* Anterosuperiorly → lateral ventricle through → foramen of Monro. (each side)

* Posteroinferiorly → fourth ventricle through → cerebral aqueduct

Boundaries :-

* **Anterior wall** - Lamina terminalis
  - Anterior commissure
  - Anterior columns of fornix

* **Posterior wall** - Pineal body
  - Posterior commissure
  - cerebral aqueduct

* **Roof** - Body of fornix
  - tela choroidea

* **floor** - Mammillary bodies
  - infundibulum (Pituitary stalk)
  - Optic chiasma
  - Tegmentum of midbrain

* **Lateral wall** - medial surface of thalamus
  - Hypothalamus

Recesses :- Recesses are extension of cavity
- suprapineal
- Pineal
- infundibular
- Optic
- Vulda

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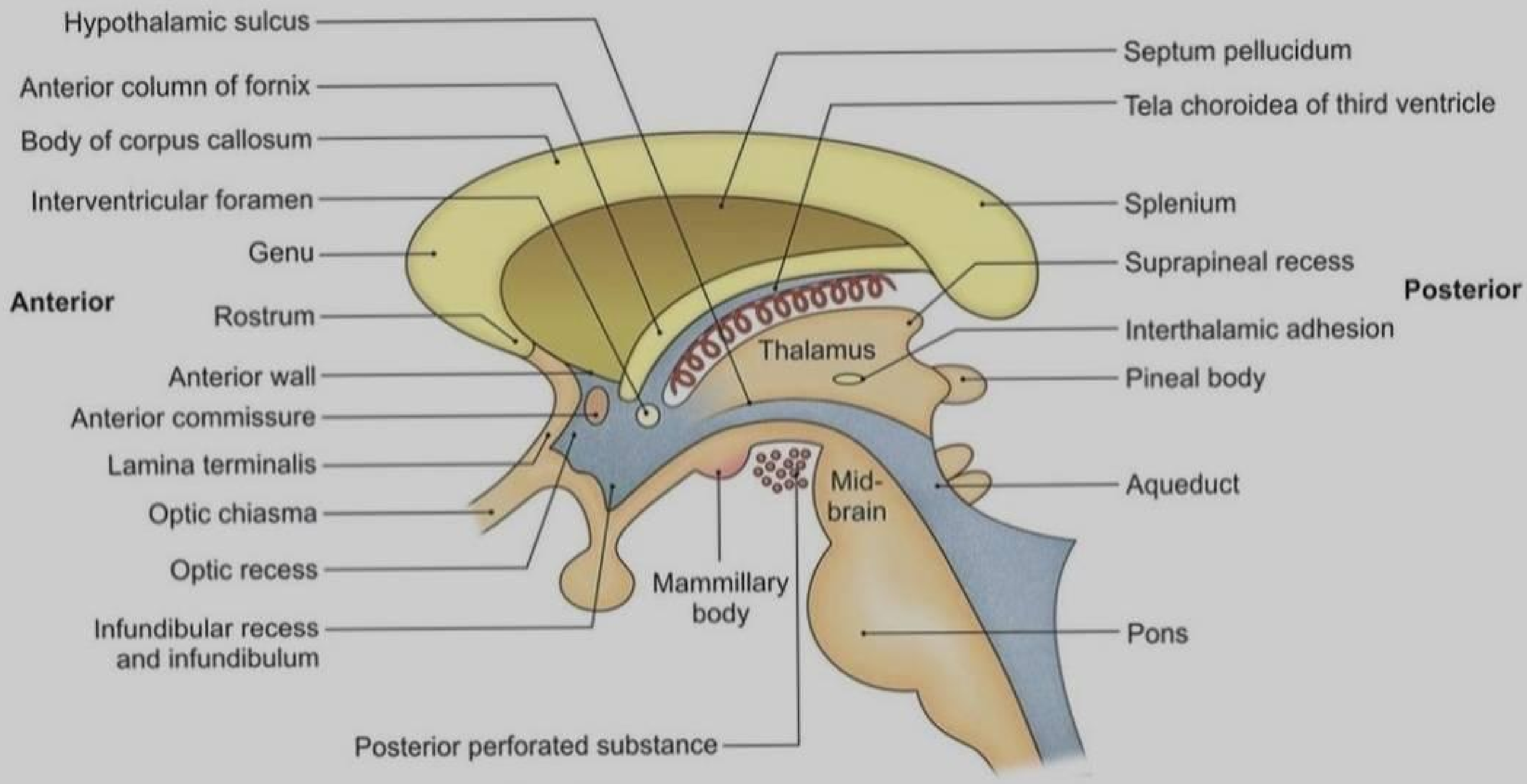


Fig. 9.1: Boundaries of third ventricle

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Fourth Ventricle :-

- cavity of hind brain
- tent-shaped space between pons and upper part of medulla oblongata

Structural layers :- floor is lined by
- ependyma
- thin layer of neuroglia
- layer of grey matter

Boundaries :-

* superolaterally → superior cerebellar peduncle
* inferolaterally → inferior cerebellar peduncle
  - gracile tubercles
  - cuneate tubercle

* floor → also called rhomboid fossa
  - formed by
    - Dorsal surface of lower pons
    - Dorsal upper of medulla oblongata

- **Part :-** It is divisible into
  - upper triangular part formed by → dorsal surface of Pons
  - lower triangular part formed by → dorsal surface of medulla
  - Intermediate part → junction of pons and medulla
    - part is marked by transversely runing fibre → stria medullaris

- **Feature of floor :-**
  - Dorsal median sulcus → divide floor → into 2x Symmetrical halves
  - sulcus limitans divide each half → median eminence, lateral eminence
  - sulcus limitans presents depression
    - at cranial end called - superior fovea
    - At caudal end called - inferior fovea

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- medial eminence → facial colliculus deep to F.C is genu of facial nerve.
- Uppermost part of sulcus limitans → bluish in colour → called locus coeruleus
  ↓ colour due to Pigmented neurons
- Descending from inferior fovea → sulcus run → toward midline oblique
  ↓ divides medial eminence into two triangle
  hypoglossal triangle
  Vagal triangle

* **Roof** :- diamond-shape and divide into → Superior inferior Part.
  - **Superior / Cranial part** form by → superior cerebellar Peduncle
    superior medullary velum
    - SCP emerging → white matter of cerebellum reach → inferior colliculi
    - SMV made up of → nervous tissue b/w SCP
  - **inferior / caudal part** form by → ventricular ependyma and
    - double flod of pia matter or tela choroidea of 4th ventricle

Communication :- → superiorly → cerebral aqueduct
→ inferiorly → central canal

openings in Roof :- 3x opening → 1x median → foramen of magendie
→ 2x - lateral → foramena of Luschka

Recesses of fourth ventricle :-
- 2x lateral recesses
- 1x median dorsal recesses
- 2x lateral dorsal recesses

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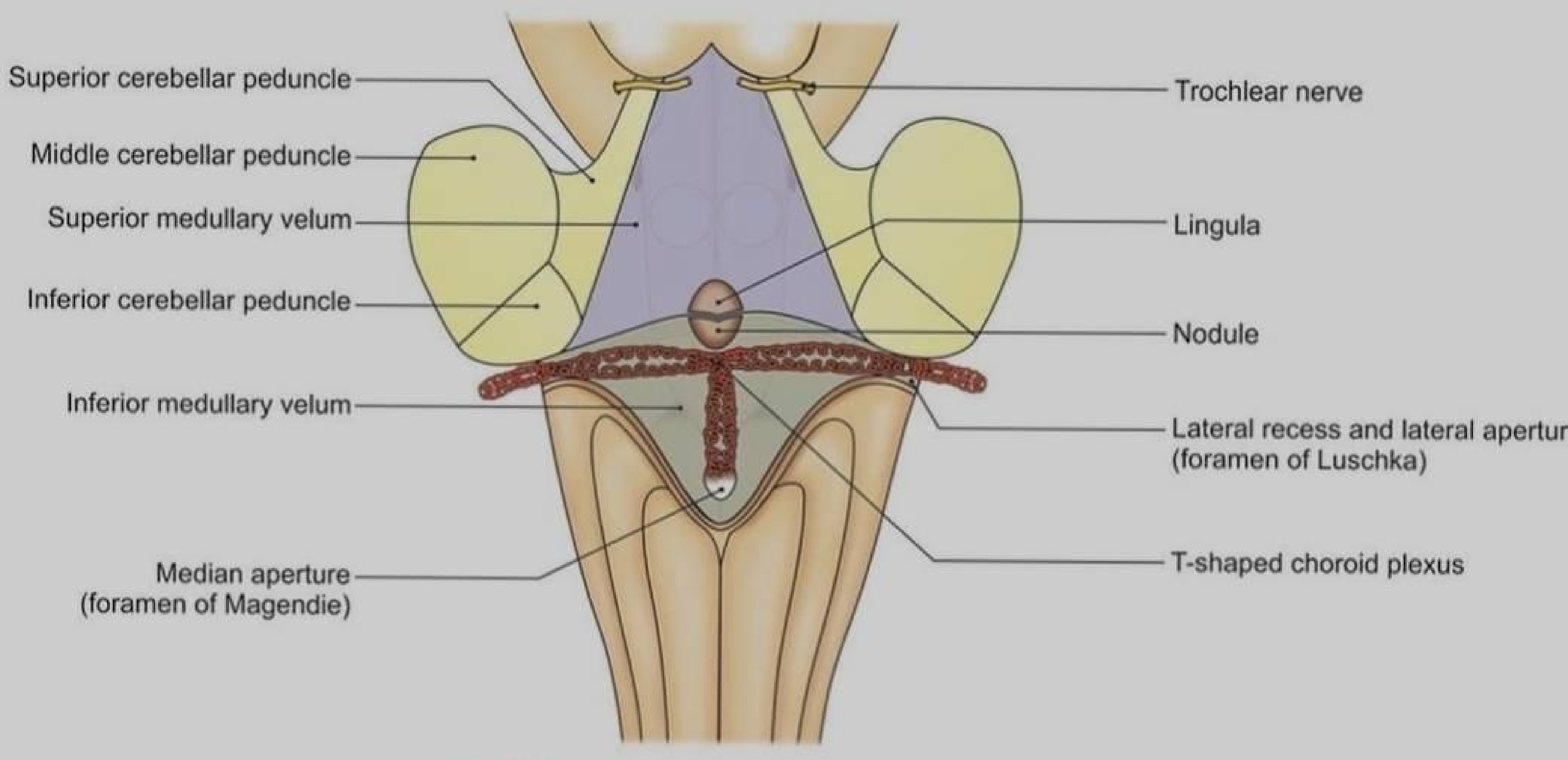


Fig. 7.2: Schematic diagram of roof of IV ventricle

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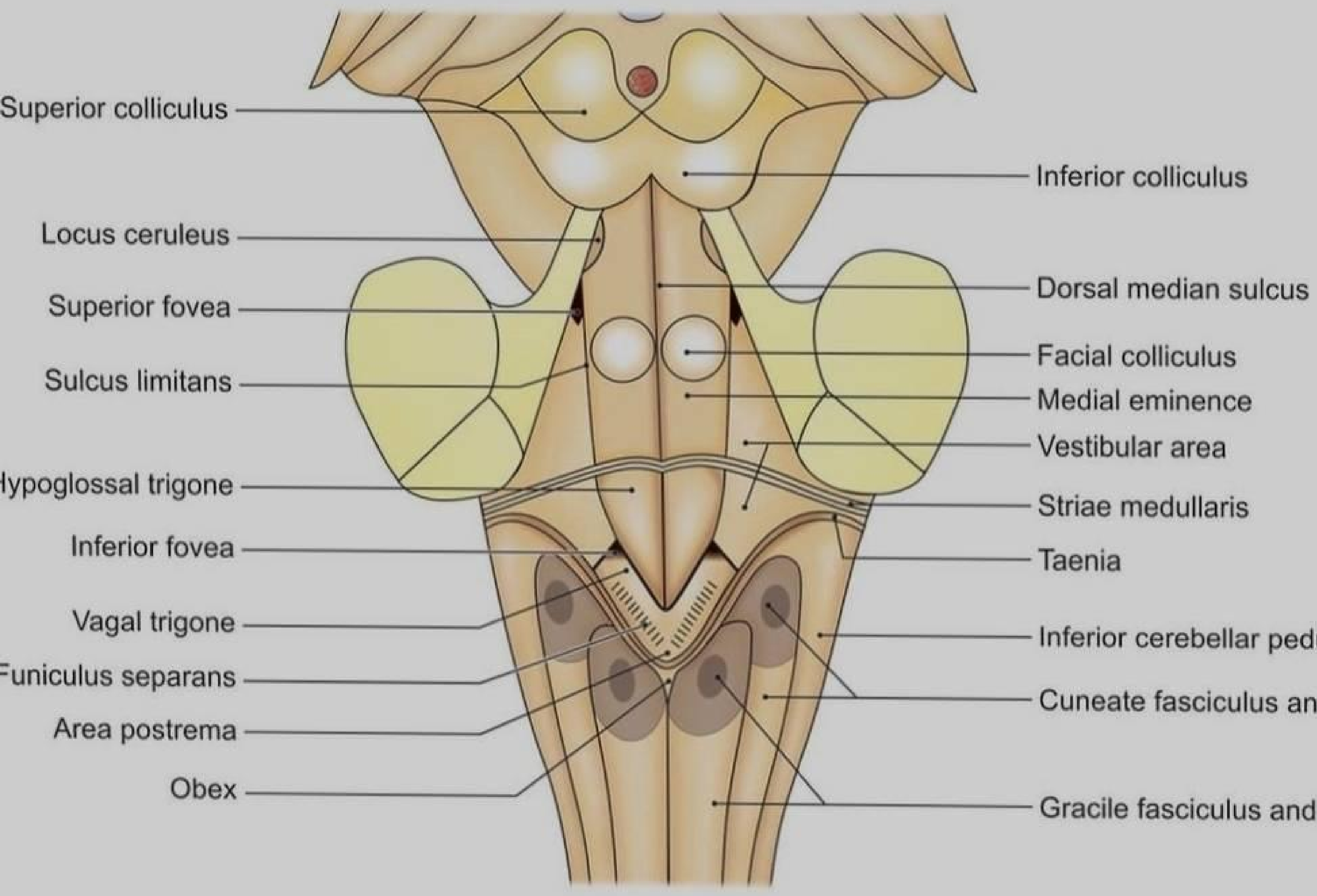


Fig. 7.1: Boundaries of IV ventricle and structures in its floor

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BASAL NUCLEI ...

- Basal nuclei are subcortical, intracerebral masses of grey matter → forming important part of Extra-pyramidal system
- includes – Caudate striatum :-
  - caudate nucleus
  - lentiform nucleus → Putamen, globus
- Amygdaloid body
- Claustrum

Corpus Striatum :-

- Caudate nucleus - C-shaped nucleus → surrounded → lateral ventricle; encloses → thalamus and internal capsule
  - head, body and tail
  - Relation - Head → form floor → Anterior horn of lateral ventricle; medial wall of anterior limb of internal capsule
  - Body → form floor → Central part of lateral ventricle; Posterior limb of internal capsule
  - Tail → form Roof → inferior horn of lateral ventricle; end by joining → amygdaloid body at temporal Pole

Lentiform nucleus - large lens shaped nucleus
- form lateral boundary of internal capsule
- 3 x surface
  - lateral surface → External capsule, Claustrum, insula
  - medial surface → internal capsule, caudate nucleus
  - inferior surface → sublentiform → separate → optic tract; tail of Caudate nucleus; inferior horn of lateral ventricle

Putamen, Globus Pallidus

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- **Amygdaloid body** - nuclear mass in temporal lobe
  - anterosuperior to inferior horn of lateral ventricle
  - Topographically → continuous with tail of caudate nucleus
  - functionally related → stria terminalis
  - function :- Emotional control.

- **Claustrum**
  - saucer-shaped nucleus
  - situated b/w → putamen and insula

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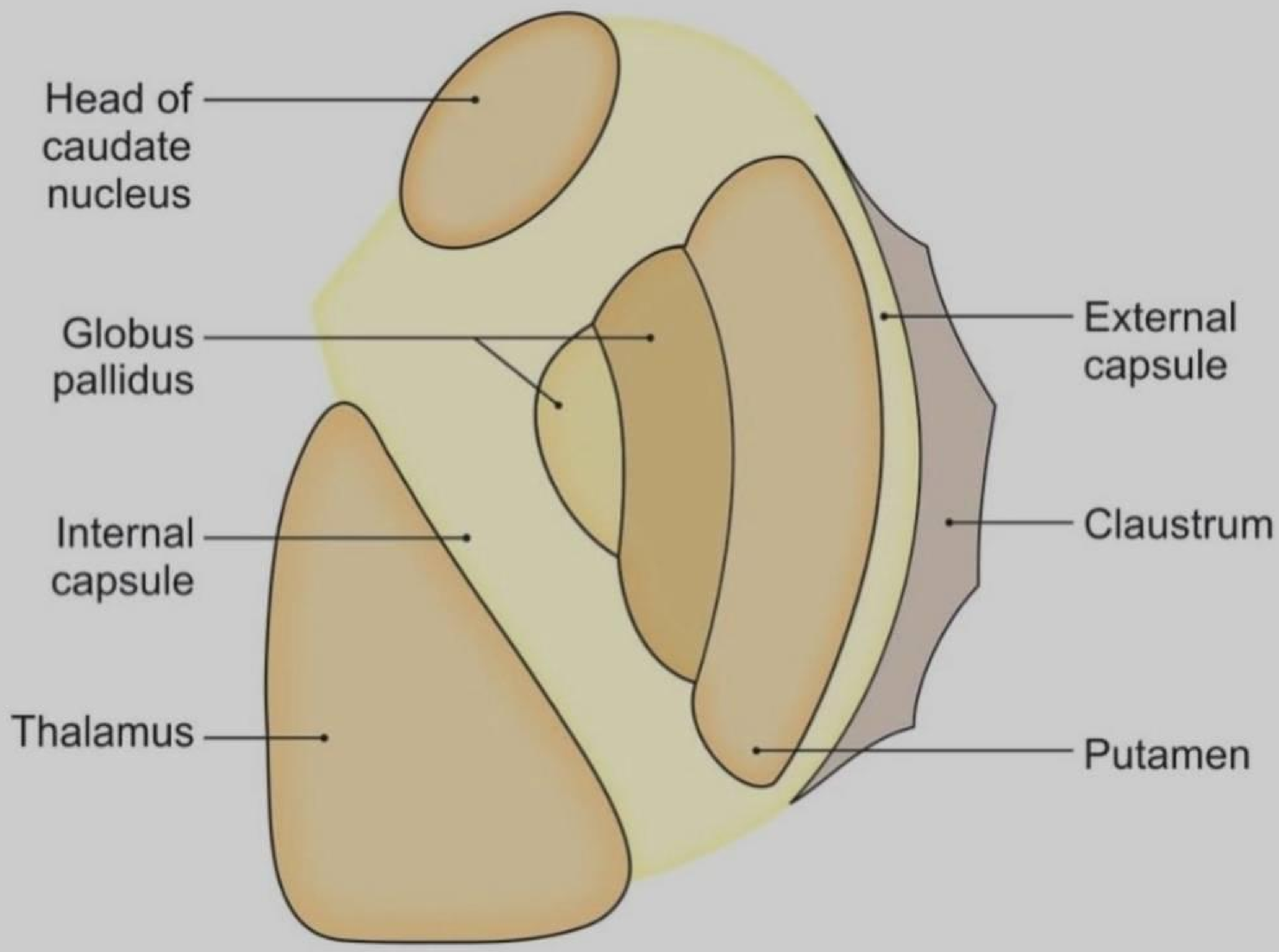


Fig. 8.20: Horizontal section through corpus striatum, thalamus and internal capsule

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Internal Capsule :-

Neuro - 2

- large band of fibres in → inferomedial part of each cerebral hemisphere
- contains fibres going to and coming from cerebral cortex
- when traced upward, → fibres of capsule diverge and continuous ↓ with corona radiata.
- when traced downward, its fibres converge and continuous ↓ with crus cerebri of midbrain

Internal Capsule divided into :-

- Anterior limb lie → between head of caudate nucleus and lentiform nucleus
- genu bend → between Anterior and posterior limb
- Posterior limb → between thalamus and lentiform nuclei
- sublentiform part → lie below lentiform nucleus
- Retrolentiform part → lie behind lentiform nucleus

Relation :-

- Medially - Head of caudate nucleus & thalamus
- Laterally - Lentiform nucleus

Fibres of Internal Capsule :-

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- Motor fibres → Corticopontine
  Frontopontine
  Parietopontine
  Temporopontine
- Pyramidal fibres → corticonuclear
  → corticospinal
- Extrapyramidal fibres → Corticorubral fibre
- Sensory fibres - Thalamocortical fibres from thalamic radiation
  - Anterior thalamic radiation
  - Superior thalamic radiation
  - inferior thalamic radiation
  - Posterior thalamic radiation

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Blood supply :-

- Anterior limb → Branch of anterior cerebral
  Artery from recurrent branch

- Genu → Branch of internal carotid
  - Branch of posterior communicating

- Posterior limb → Branch of anterior choroidal
  - Branch of posterior communicating
  - Branch of lateral and medial striate arteries

- Sublentiform → Branch of anterior choroidal
  Branch of posterior cerebral

- Retrolentiform → Branch of posterior cerebral

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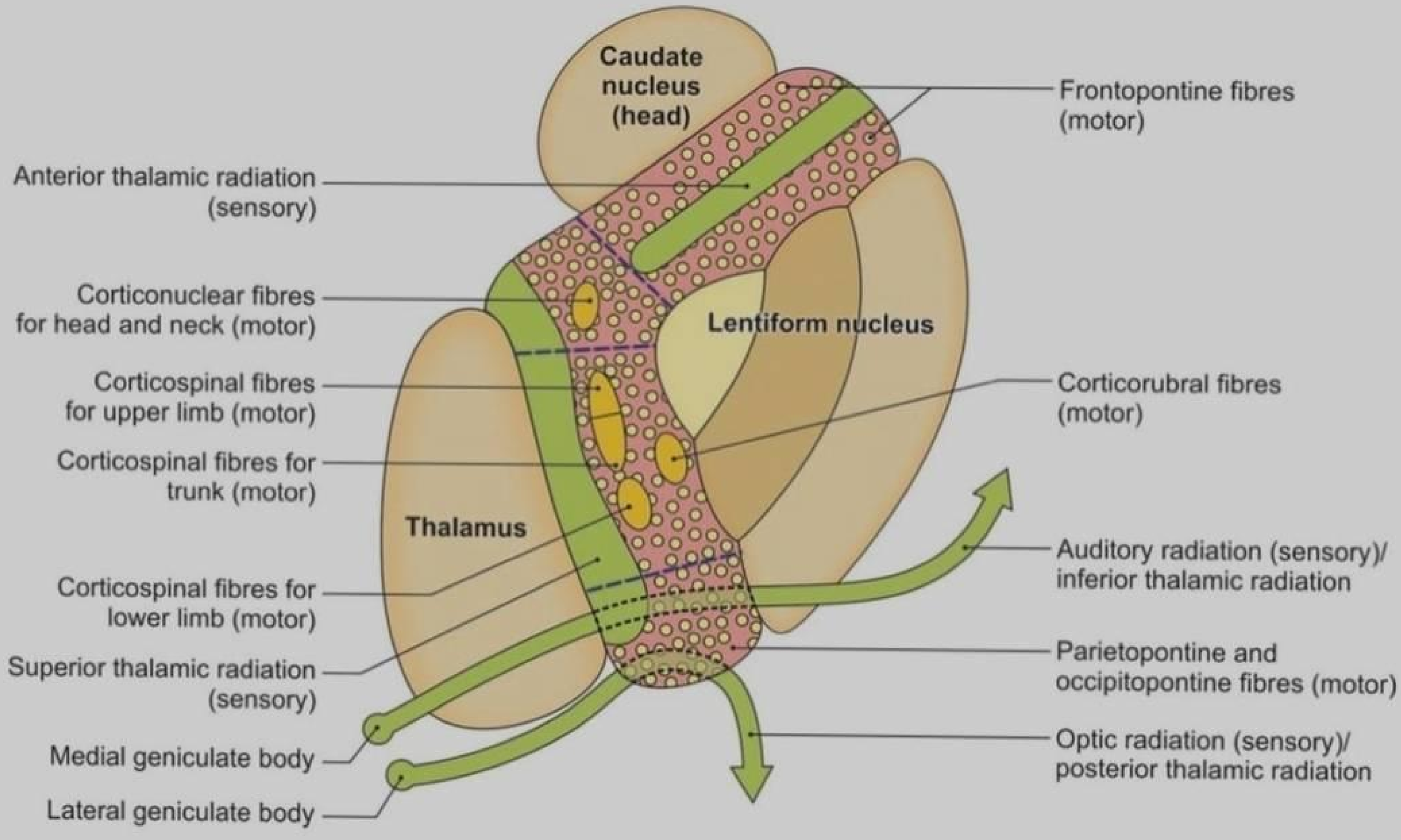


Fig. 8.29: Fibre components of internal capsule

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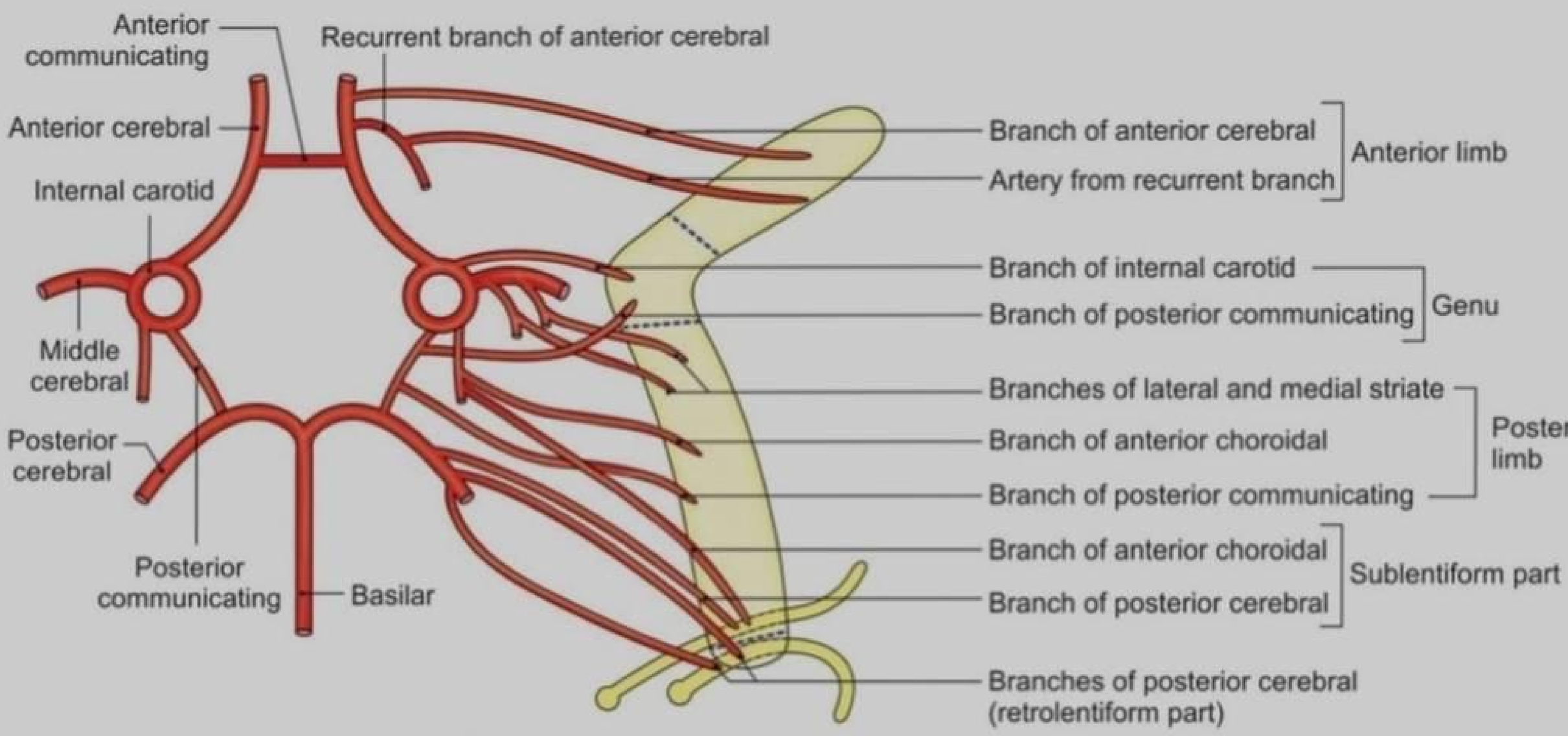


Fig. 8.30: Arteries supplying the internal capsule

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CORPUS CALLOSUM ...

- largest commissure of the brain.
- connect 2x cerebral hemisphere

anterior commissure ← which connected by lower and anterior part of temporal lobes except

Part of brain connected :-

- connects all part of cerebral cortex of 2x side

Part of Corpus Callosum :-

* **Genu** - 4cm behind the frontal pole
  - Relation - anteriorly → anterior cerebral arteries
    Posteriorly → Anterior horn of → lateral ventricle

* **Rostrum** - directed- downward and backward from genu
  - end by joining → lamina terminalis in front of Anterior commissure
  - Relation - superior to Anterior horn

* **Trunk** - Middle part, between genu and Rostrum
  - overlapped by → cingulate gyrus.

* **Splenium** - 6cm front of occipital pole
  - Relation → inferior surface - tela choroidea
    superior surface - inferior sagittal sinus

* **Fibres of Corpus Callosum :-**
  * Rostrum connect → orbital surface of 2x frontal lobes
  * forceps minor connect → 2x frontal lobes (made up of fibres of genu)
  * forceps major connect → 2x occipital lobes (made up of fibres of splenium)
  * tapetum → (formed by - some fibres from trunk & splenium of corpus callosum
    - It forms → roof and lateral wall of posterior horn
      → lateral wall of inferior horn

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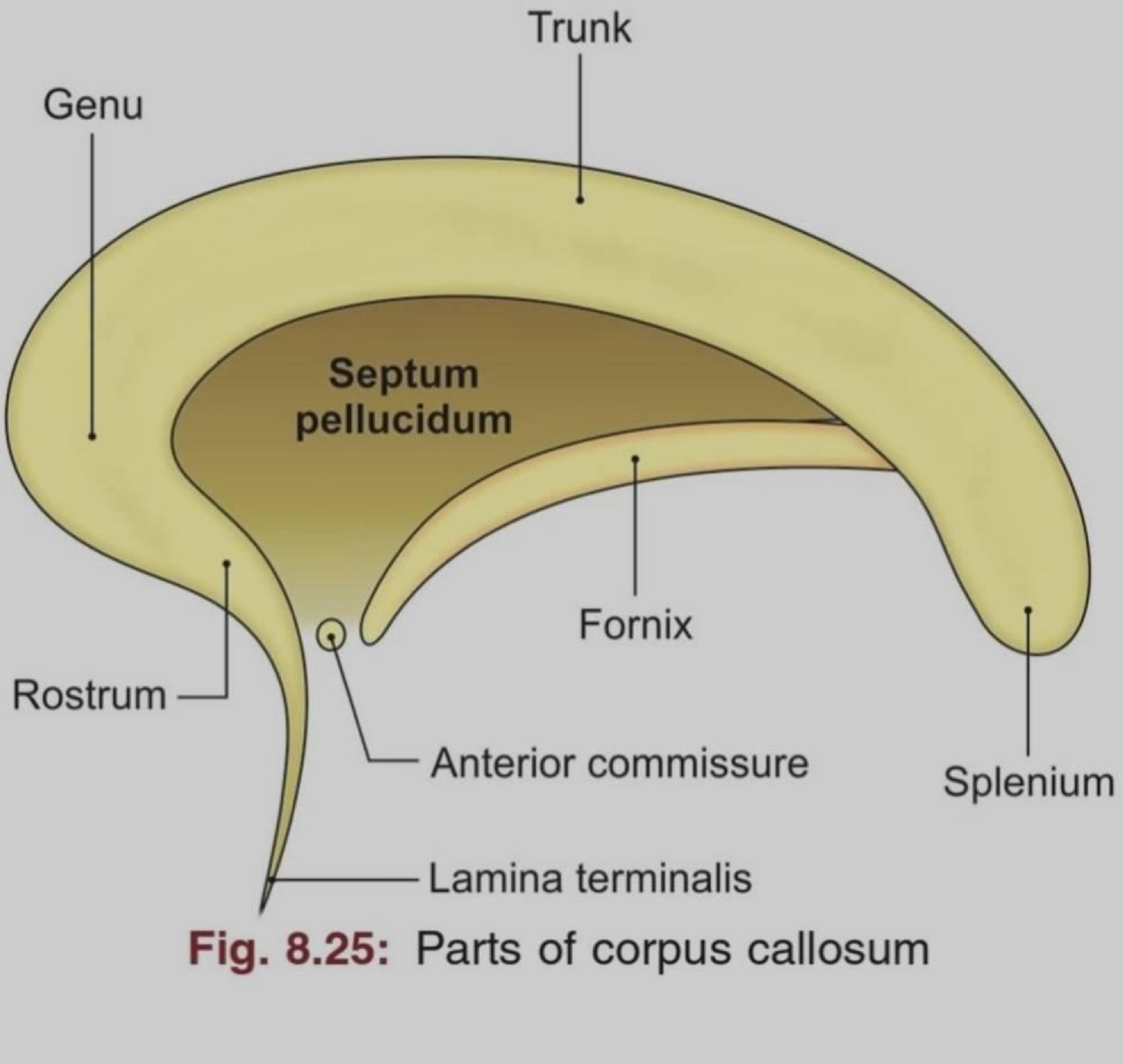


Fig. 8.25: Parts of corpus callosum

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White Matter OF Cerebrum ...

- White matter of cerebrum consist of myelinated fibres
- which connect various part of cortex to one another
- fibres are classified into 3x group – Association fibres
  - commissural fibres
  - Projection fibres

*Association fibres (Arcuate) :-

- Connect different cortical area of same hemisphere to one another
- subdivided into 2x group

1) Short association fibres - connect → adjacent gyri

2) long association fibres connect → more widely separated gyri

- uncinate fasciculus → connect temporal lobe to → motor speech area
  → orbital cortex

- Cingulum → connect cingulate gyrus to → Parahippocampal gyrus

- superior longitudinal fasciculus → connect frontal lobe to → occipital lobe
  → temporal lobes

- inferior longitudinal fasciculus → connect occipital lobe to → temporal lobe

Commissural fibres :- connect → corresponding part of 2x hemisphere

- Corpus calloscum - connecting → cerebral cortex of two side
- Anterior commissure - → archipallia of both side
- Posterior commissure - → superior colliculi
- commissure of fornix → connecting the crura of fornix

Internal capsule fibres :-

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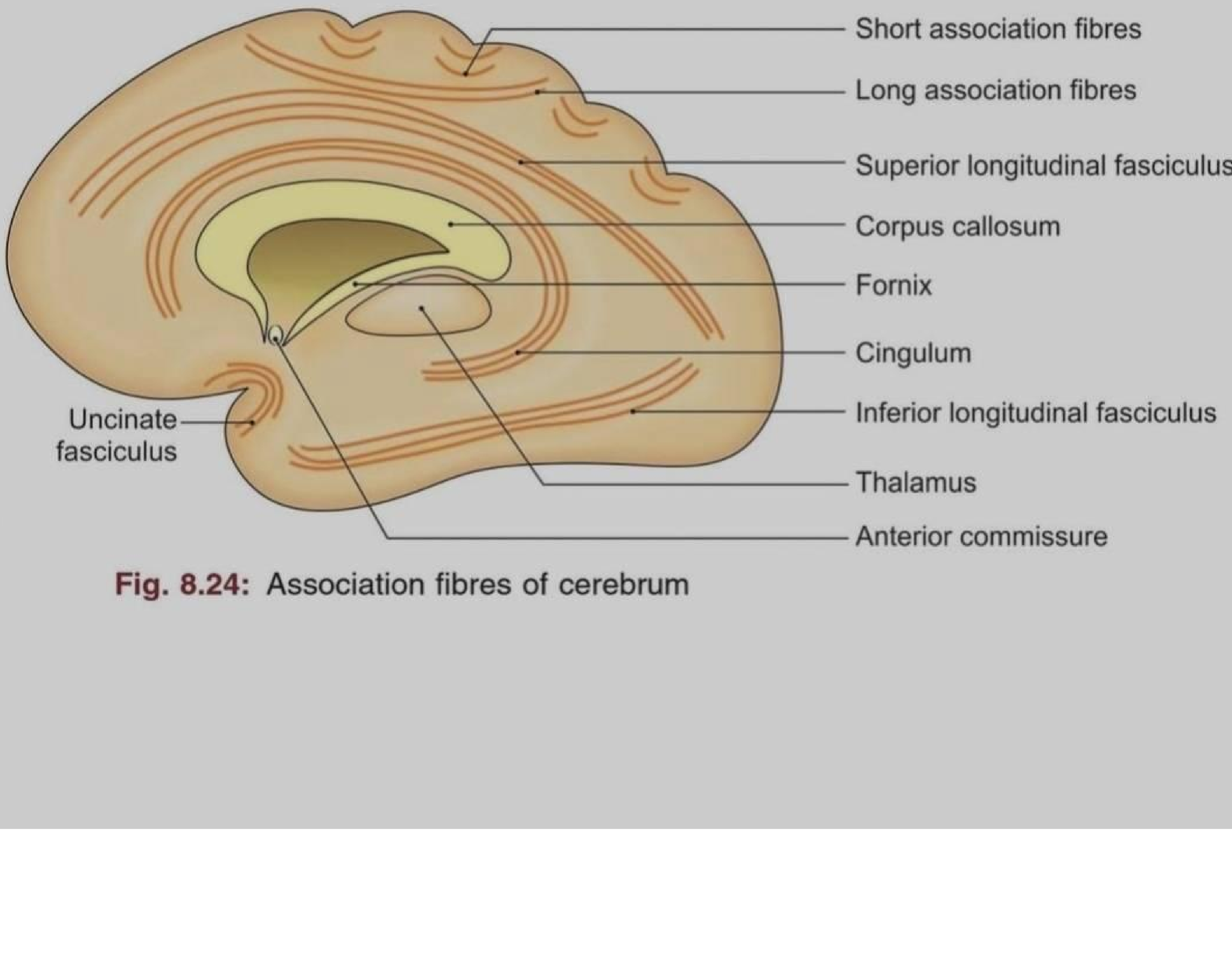


Fig. 8.24: Association fibres of cerebrum

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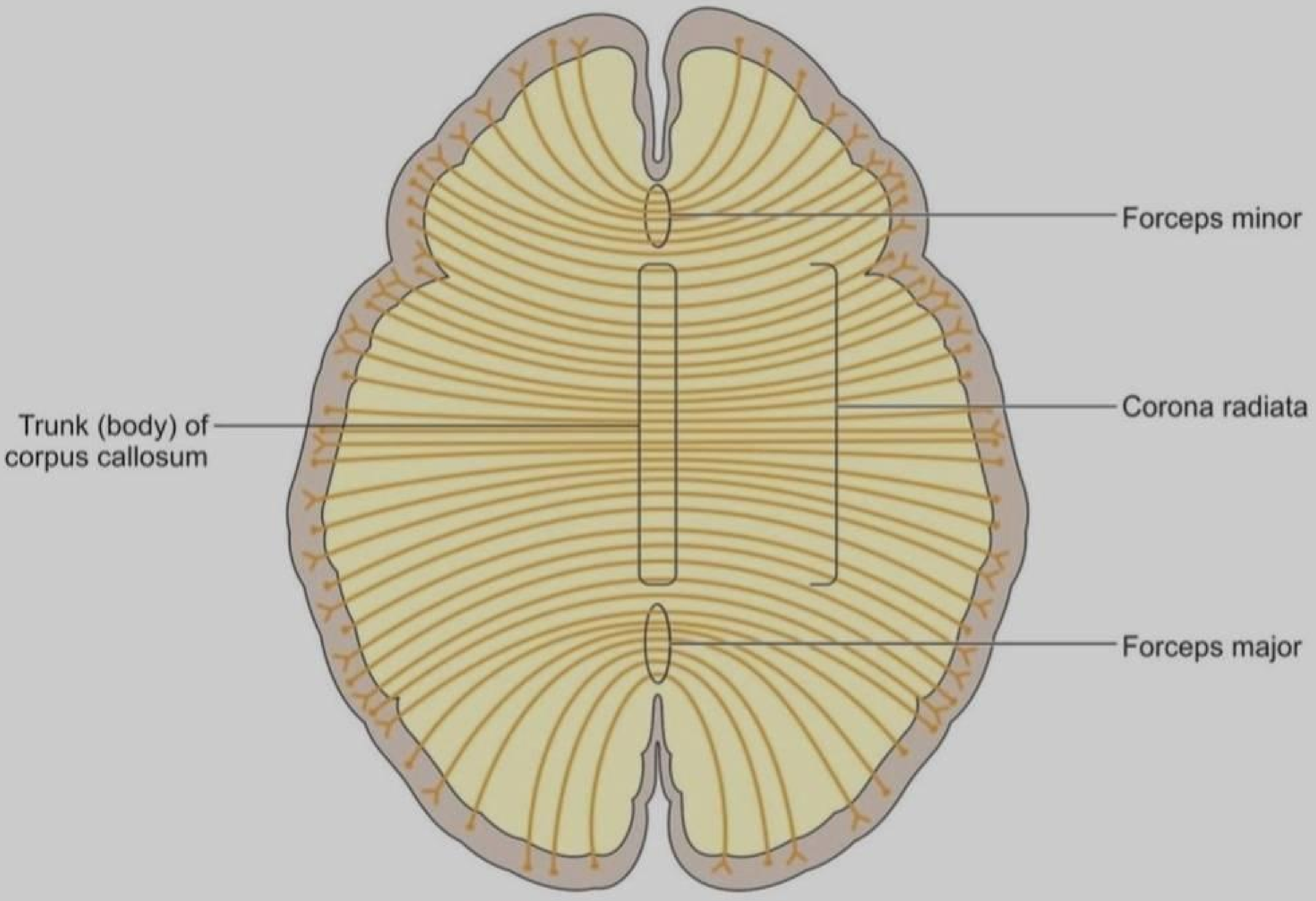


Fig. 8.26: Fibres of corpus callosum

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CISTERNS OF SUB-ARACHNOID SPACE ...

- At base of brain and around brainstem → subarachnoid space forms → intercommunicating Pools ↓ called Cisterns
- Reinforce the protective effect of CSF
- Cerebellomedullary cistern :-
  - lying in the angle b/w medulla, cerebellum and occipital bone
  - receives → CSF from → 4th ventricle via → median foramen of Magendie, foramina of Luschka
  - contain - Vertebral artery
    - cranial nerve - 9th, 10, 11, 12
    - choroid plexus

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- Cisterna Pontis :-
  - It is present on → ventral aspect of pons
  - continuous with → interpeduncular cistern
  - contain - Basilar artery
    - Superior cerebellar artery
    - sixth (VI) cranial nerve

- Interpeduncular Cistern :-
  - large cistern b/w two temporal lobe
  - continous with subarachnoid space around → anterior, middle, Posterior cerebral A.
  - contain - Basilar artery
    - Penduncular segment of PICA, SCA
    - Perforating branch of PICA (Posterior inferior cerebellar artery)
    - optic chiasma
    - 3rd cranial nerve

- Cistern of great cerebral vein :-
  - lie in space between splenium of corpus callosum and superior surface of cerebellum
  - contain - Pineal gland, great cerebral vein

- Lumbar Cistern - large sub-arachnoid space in lumbar region ↓ distal to L3 - L4

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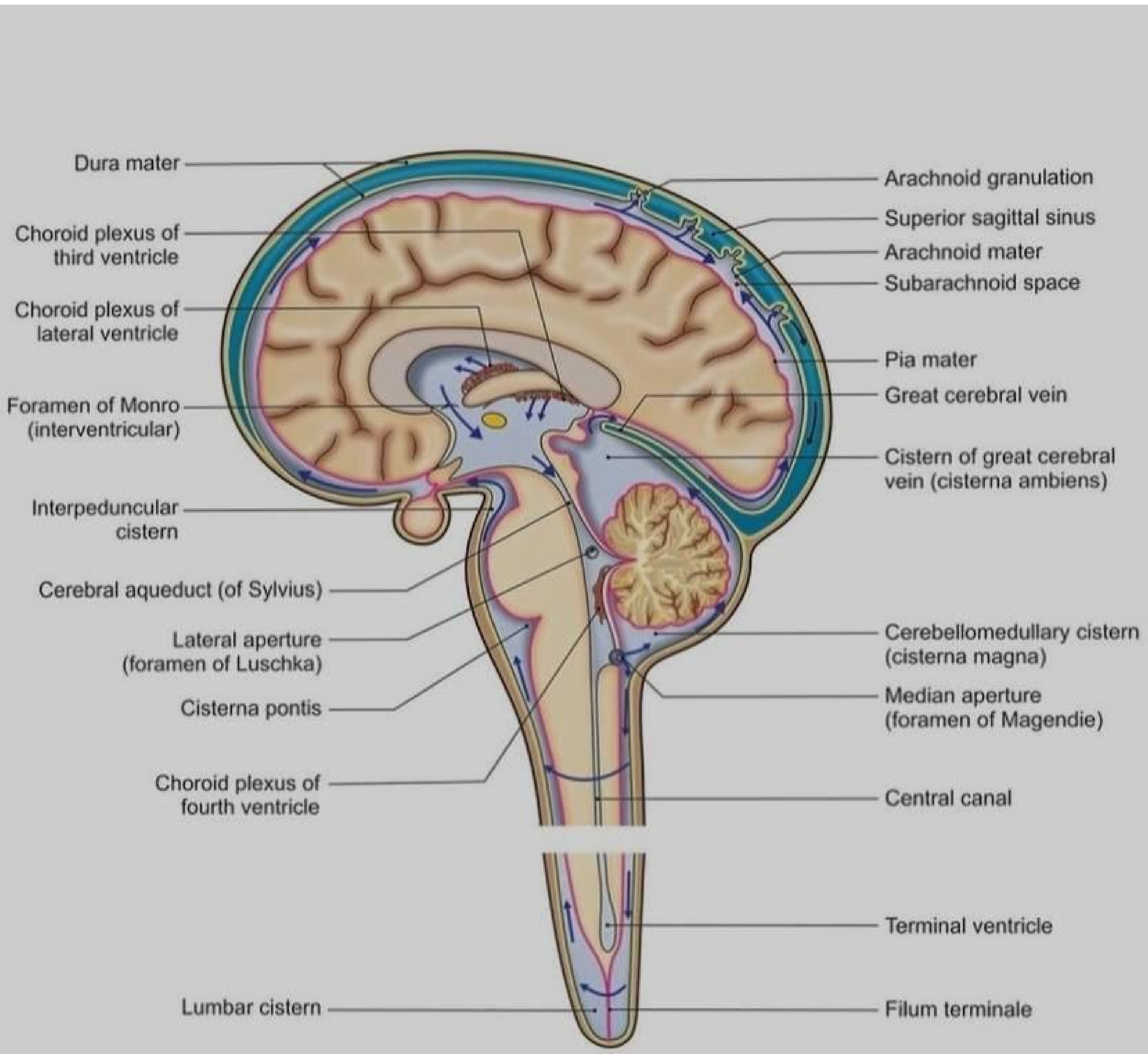


Fig. 2.6a: Various cisterns of the brain including formation, circulation and absorption of cerebrospinal fluid. Arrows show the direction of flow of CSF

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FACIAL NERVE...

neuro-anatomy

- Facial nerve is nerve of → second branchial arch

Nuclei :- fibres of nerve are connected to 4x nuclei → situated in lower pons
- motor nucleus
- superior salivatory nucleus or Parasympathetic
- Lacrimatory nuclei – Parasympathetic
- Nucleus of tractus solitarius

Course :-
- Attached to brainstem by 2 roots, motor and sensory
- two roots of facial nerve are attached to → lateral part of pons
- Two roots run laterally and forward with → 8th cranial nerve ↓ to reach internal acoustic meatus
- In meatus :-
  - ↳ At bottom of meatus → 2x roots, sensory and motor fuse ↓ to form single trunk ↓ lie in Petrous of temporal lobe
  - → with in meatus → divided into → 3x parts
    - 1st part → directly laterally above vestibule
    - 2nd part → related to medial wall of middle ear above → Promontory.
    - 3rd part → directly vertically downward behind → Promontory
- facial nerve leave skull by passing through → stylomastoid foramen
- **Extracranial course :-**
  - facial nerve crosses the → lateral side of Base of styloid Process
  - It enters – posteromedial surface of parotid gland
  - runs forward through the gland crossing → retromandibular vein and external carotid A.
  - Behind the neck of mandible, it divide into 5x terminal branches ↓ emerge along anterior border of Parotid gland

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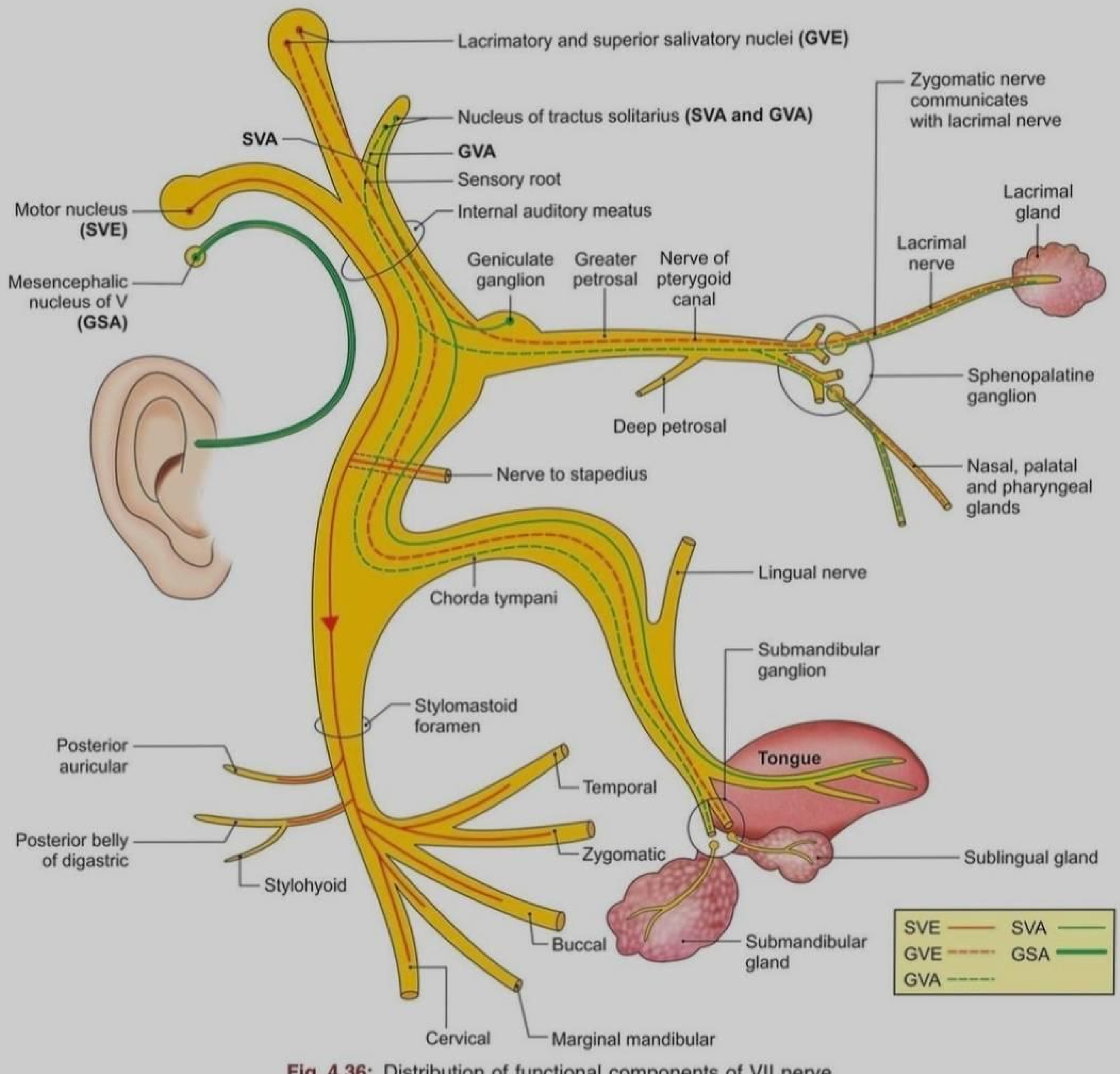


Fig. 4.36: Distribution of functional components of VII nerve

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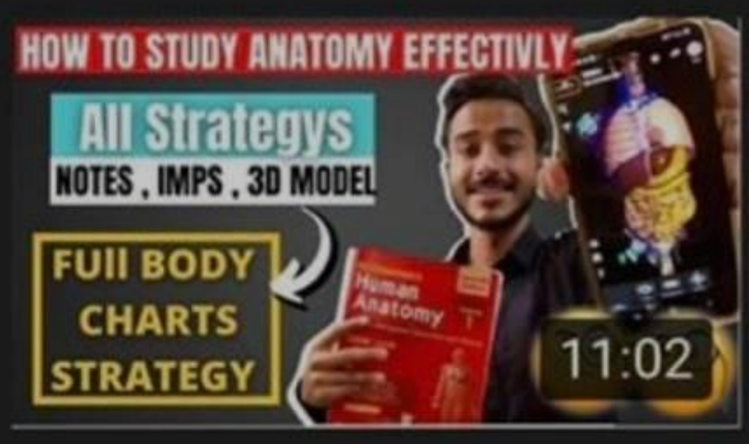
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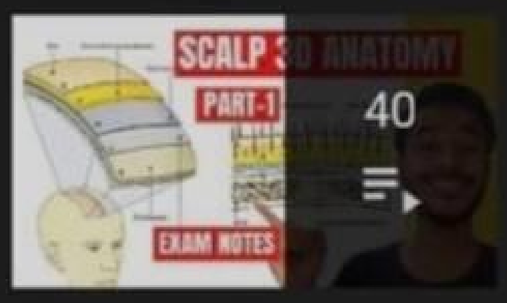
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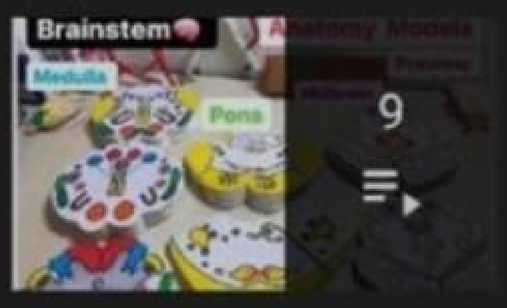
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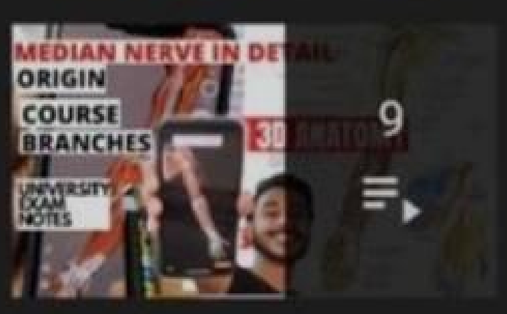
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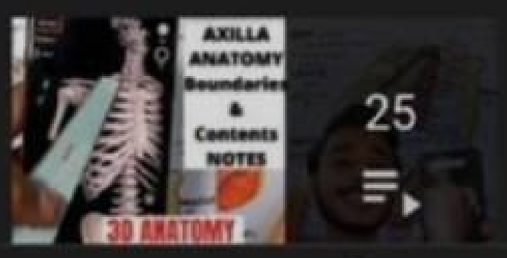
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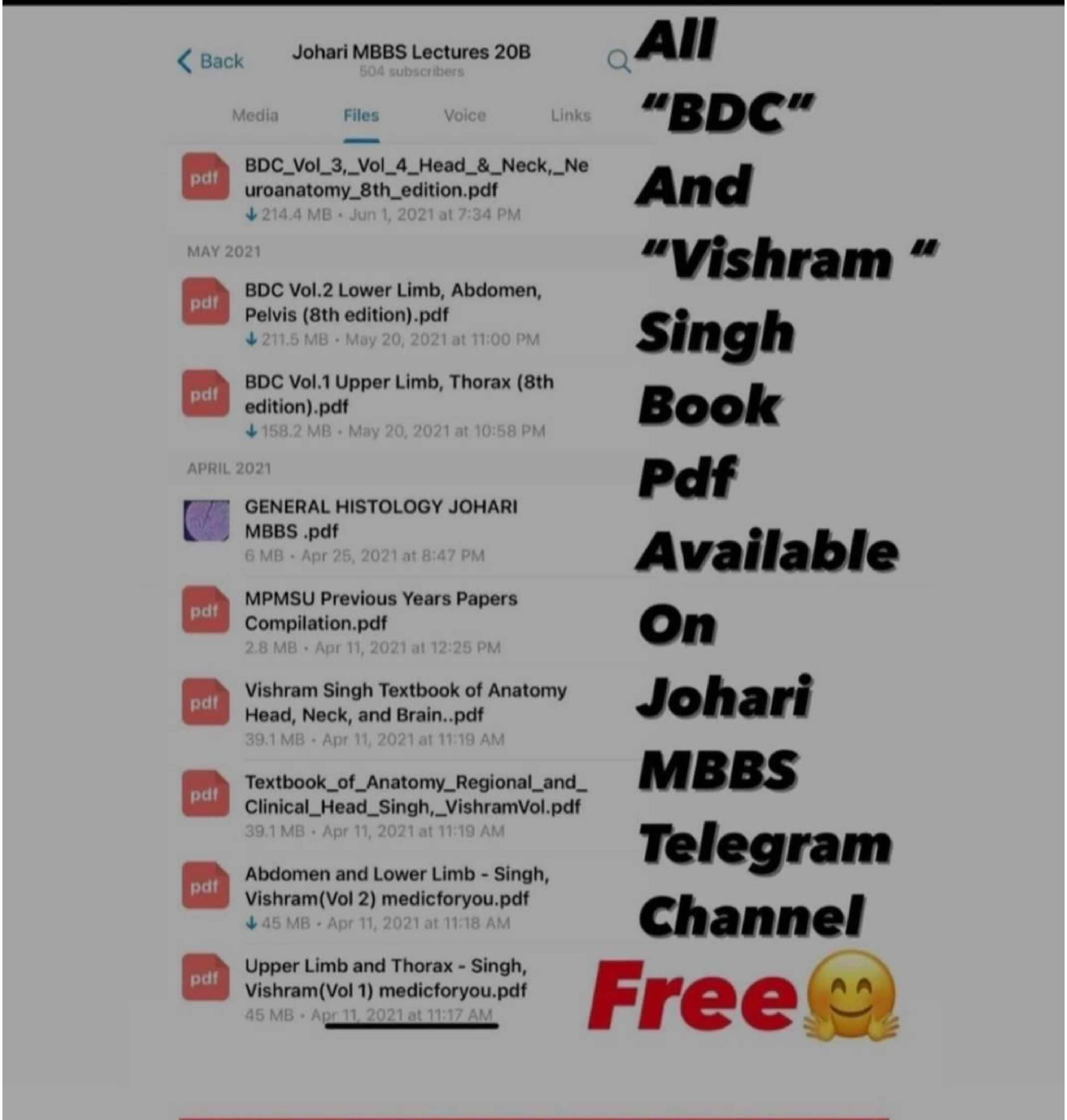


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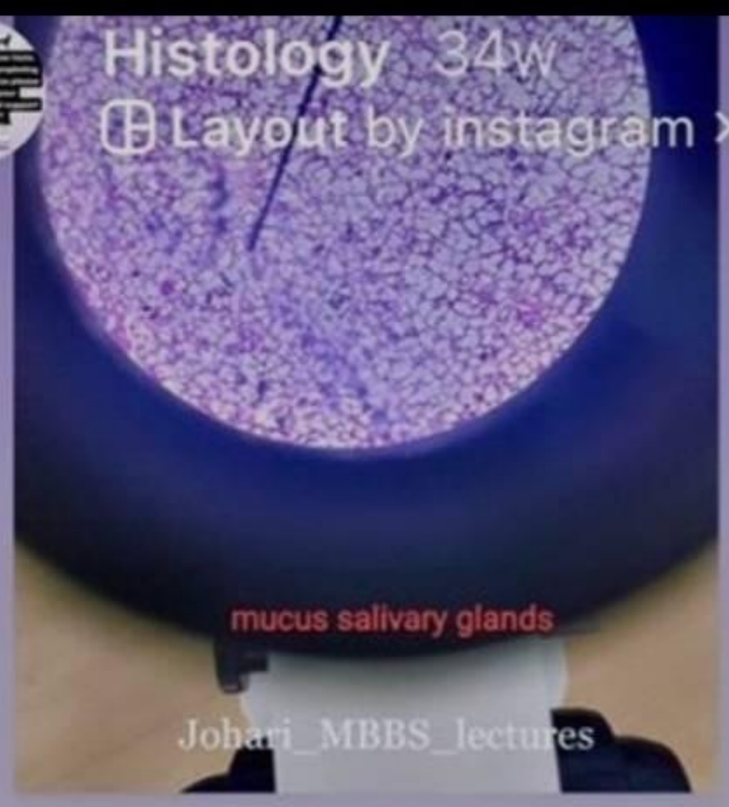
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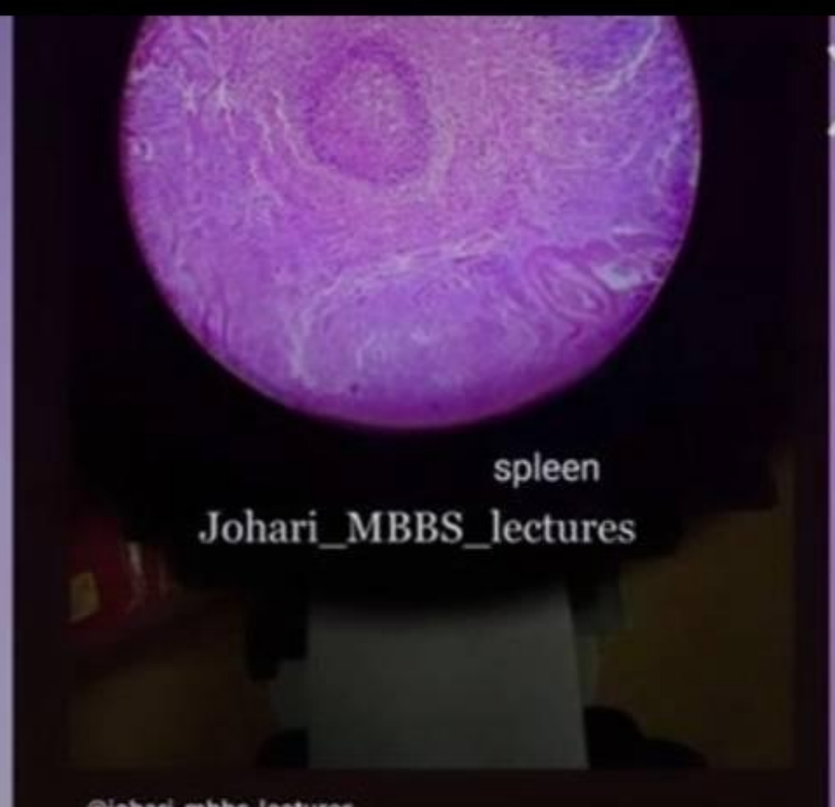
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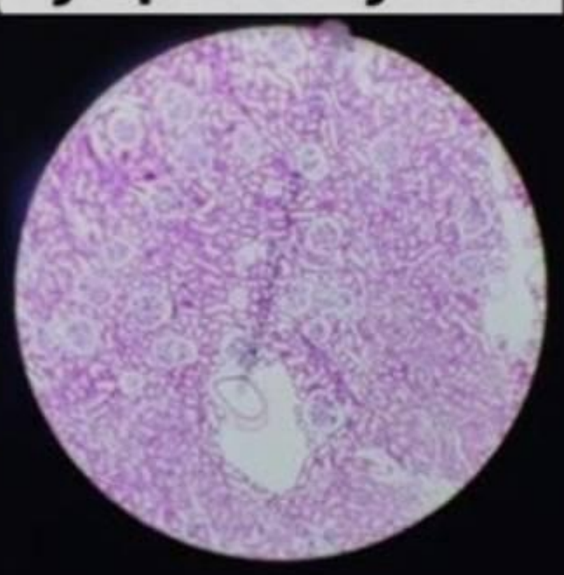


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Lymphatic system

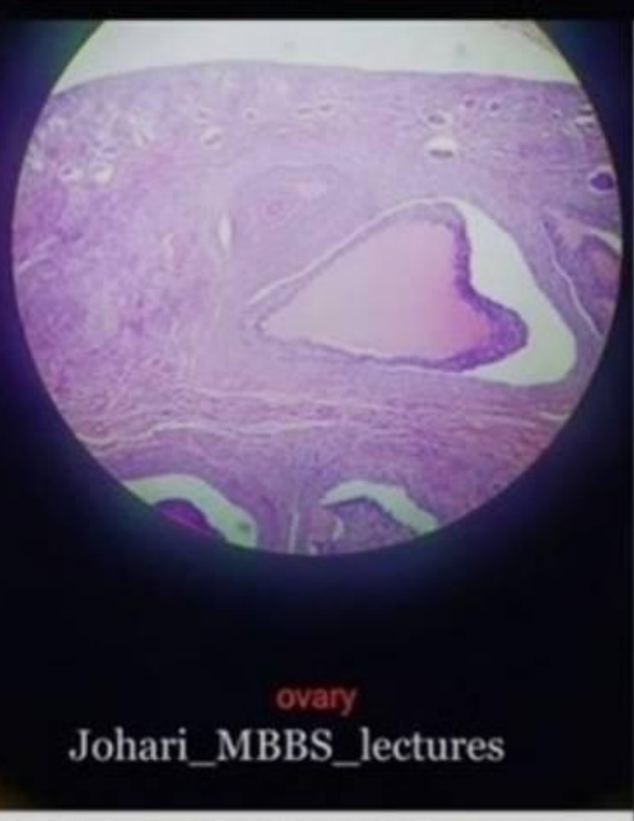


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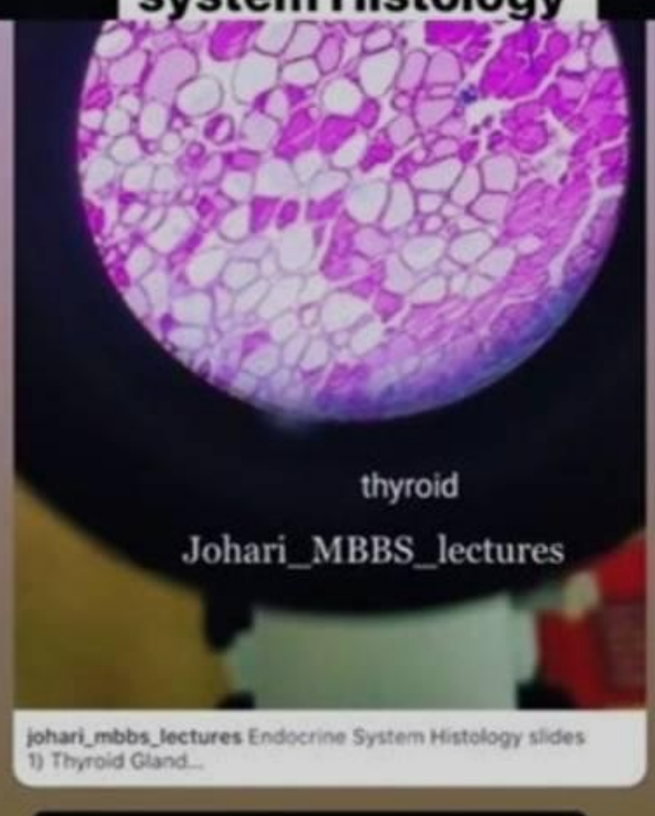
GIT Histology slides



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Reproductive system

Urinary & Respirative system Histology



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Endocrine system Histology slides

ALL SYSTEMS HISTOLOGY SLIDES

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